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Antimicrobial peptides in nematode secretions – Unveiling biotechnological opportunities for therapeutics and beyond

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ABSTRACT

Gastrointestinal (GI) parasitic nematodes threaten food security and affect human health and animal welfare globally. Current anthelmintics for use in humans and livestock are challenged by continuous re-infections and the emergence and spread of multidrug resistance, underscoring an urgent need to identify novel control targets for therapeutic exploitation. Recent evidence has highlighted the occurrence of complex interplay between GI parasitic nematodes of humans and livestock and the resident host gut microbiota. Antimicrobial peptides (AMPs) found within nematode biofluids have emerged as potential effectors of these interactions. This review delves into the occurrence, structure, and function of nematode AMPs, highlighting their potential as targets for drug discovery and development. We argue that an integrated approach combining advanced analytical techniques, scalable production methods, and innovative experimental models is needed to unlock the full potential of nematode AMPs and pave the way for the discovery and development of sustainable parasite control strategies.

List of abbreviations

Antimicrobial Peptide	AMP
Clustered Regularly Interspaced Short Palindromic Repeats	CRISPR
Colony Forming Units	CFU
Excretory-Secretory Products	ESPs
Gastrointestinal	GI
Liquid Chromatography with Tandem Mass Spectrometry	LC-MS/MS
Parasitic Gastroenteritis	PGE
Parallel Reaction Monitoring	PRM
Post-Translational Modifications	PTM
Pseudocoelomic Fluid	PCF
RNA Interference	RNAi
Trapped Ion Mobility Spectrometry with Time-of-Flight	TimsTOF

1. Introduction

Gastrointestinal (GI) parasitic nematodes affect human and animal health worldwide. Collectively, soil-transmitted parasitic nematodes of humans [*Ascaris* spp., hookworms (*Ancylostoma duodenale* and *Necator americanus*), and whipworms (*Trichuris trichiura*)], are estimated to infect almost one billion people worldwide, particularly in the poorest regions of Africa, Southeast Asia, and South America (reviewed by Loukas et al., 2021). These infections cause a range of morbidities including chronic intestinal disease, hepatic disorders and severe anaemia (reviewed by Loukas et al., 2016; Else et al., 2020). Similarly, the aetiological agents of livestock parasitic gastroenteritis (PGE) such as the 'barber's pole worm', *Haemonchus contortus*, the 'brown stomach worms', *Teladorsagia circumcincta* and *Ostertagia ostertagi*, and the 'black scour worms', *Trichostrongylus* spp., are responsible for significant economic losses worldwide, estimated at €1.8 billion in Europe alone

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(Charlier et al., 2020). Clinical signs of PGE include diarrhoea, loss of appetite, failure to thrive and, in severe cases, death, with costs associated with stock mortality and replacement, decreases in carcass weight and changes in the composition and quality of wool, as well as reduced fertility and milk yield (reviewed by Tariq, 2015). In addition to impacts on animal health and welfare, PGE represents a significant threat to global food security (United Nations, 2024), and projected Net Zero targets (Kyriazakis et al., 2024).

Control of GI nematode infections currently relies on the implementation and delivery of mass drug administration programmes in humans, and targeted selective treatments in livestock (reviewed by Nixon et al., 2020; Gilleard et al., 2021). However, in endemic areas where GI nematodes of humans are widespread, re-infections are common (Jia et al., 2012), and the indiscriminate use of parasiticides in livestock has accelerated the global emergence of multidrug-resistant nematode populations (Charlier et al., 2022). Despite long term research efforts focussed on the development of vaccines for both nematodes of humans and livestock (*e.g.* Goud et al., 2005; Nisbet et al., 2019), there has been limited progress in this area (Nisbet et al., 2016; Wong et al., 2023; Puchner et al., 2024). Collectively, parasite control challenges underscore the importance of prioritising the discovery of new, sustainable strategies for infection management. The development of such strategies relies on an improved understanding of nematode biology and host-parasite interactions. Over recent decades, a wealth of knowledge has been gained on several aspects of host-parasite relationships, including parasite and host immune-molecular responses to infection, mechanisms of host invasion and infection establishment, and strategies of immune evasion and suppression (reviewed by Sorobetea et al., 2018; Maizels, 2020; Gilleard et al., 2021). Furthermore, recent evidence indicates the occurrence of a complex interplay between GI parasitic nematodes of humans and livestock and the host gut microbiota (reviewed by Peachey et al., 2017; Cortés et al., 2020; Rooney et al., 2022a). In particular, several studies involving parasitic nematodes of humans [*e.g. N. americanus* (Ducarmon et al., 2020) and *Strongyloides stercoralis* (Jenkins et al., 2018)] and livestock [*e.g. T. circumcincta* (Cortés et al., 2020; Rooney et al., 2022b) and *H. contortus* (Li et al., 2016)], conducted under both experimental and natural conditions of infection, have revealed a plethora of colonisation-associated changes in host microbiota composition and function (reviewed by Cortés et al., 2019; Kupritz et al., 2021). However, the precise nature and causal relationships underlying microbiota changes associated with parasitic nematode infections remain elusive, with variable data relating to alterations in bacterial richness and diversity following parasite establishment (reviewed by Cortés et al., 2019).

A better understanding of the crosstalk between nematode parasites and the host environment will reveal the fundamental mechanisms underpinning this interplay and may inform the development of novel, sustainable strategies of pathogen control (Midha et al., 2017; Cortés et al., 2020). While these mechanisms are most likely multifactorial, involving both direct and indirect (immune-mediated) means of communication (reviewed by Peachey et al., 2017; Rooney et al., 2022a), recent studies have highlighted nematode biofluids as likely effectors of host-parasite-microbiota interactions (Rooney et al., 2024). Nematode biofluids are represented by internal body cavity fluid [pseudocoelomic fluid (PCF)] and excretory-secretory products (ESPs). The PCF and/or ESP proteomes of several GI parasitic nematodes have been characterised [*e.g. N. americanus* (Logan et al., 2020), *Ancylostoma ceylanicum* (Uzoechi et al., 2023), *Ascaris suum* (Chehayeb et al., 2014), and *T. circumcincta* (Smith et al., 2009)]. More recent studies have shown that nematode biofluids contain extracellular vesicles (EVs) rich of biologically active molecules (reviewed by Rooney et al., 2022a; Stryiński et al., 2025) that may have important endogenous and exogenous roles in nematode innate immunity (PCF), communication with the host immune system (ESPs), and host-gut microbiota interactions (ESPs) (reviewed by Rooney et al., 2022a). Indeed, the rodent nematode *Heligmosomoides polygyrus* produces ESPs with antimicrobial activity against *Enterococcus faecium* and *Escherichia coli* (Rausch et al., 2018), while ESPs of *T. circumcincta* have been shown to significantly reduce growth of *E. coli in vitro* (Rooney et al., 2022b).

The antimicrobial properties of nematode biofluids have been associated with the activity of antimicrobial peptides (AMPs). AMPs are a group of small bioactive proteins produced by a wealth of living organisms, and that are characterised by potent and broad-spectrum antimicrobial activity against multiple classes of bacteria, yeasts, fungi, viruses, and parasites (Huan et al., 2020). In nematodes, AMPs have been identified in *A. suum* PCF (Midha et al., 2018; Atkinson et al., 2021; Irvine et al., 2023a), and *T. circumcincta* ESPs (Rooney et al., 2022b), although the specific mechanisms underlying the role of worm-derived AMPs in parasite-microbiota interactions are yet to be elucidated (Irvine et al., 2023a, 2023b). Nevertheless, this area of research is attracting growing attention, particularly as nematode innate immunity may represent a source of novel antimicrobials and anthelmintics.

Here, we (i) examine the current state of knowledge of AMP protein families across key parasitic nematodes of human and veterinary significance, (ii) discuss solutions to bridge the gap between fundamental and translational research in nematode AMPs, and (iii) propose methods to harness the therapeutic potential of worm-derived AMPs to improve human and animal health.

2. Nematode AMP discovery

Throughout their life cycles, nematodes inhabit heterogeneous environments where they may be exposed to a range of host and environmental microbiota. In the absence of an adaptive immune system, the ability of nematodes to survive in microbe-rich, hostile locations is likely associated with a well-developed innate immune system that includes AMPs (Tran and Luallen, 2024). Whilst AMP profiles have been well characterised across many invertebrate species (Mylonakis et al., 2016; Guryanova et al., 2023), knowledge of nematode-derived AMPs has only recently begun to expand.

AMP discovery has evolved considerably over the last four decades; indeed, traditional strategies relied on time-consuming and costly biochemical isolation of bioactive compounds from whole animals and tissues with a focus on 'tractable' biological systems (Wang, 2020). As a result, the majority of available AMP data are derived from amphibians and arthropods (Wang et al., 2022a). More recently, the development of improved sequencing platforms and advances in the availability of -omics datasets have led to the identification and characterisation of AMPs from a broader range of invertebrate species on an unprecedented scale (Porto et al., 2017). Critically, the integration of algorithms and machine learning technologies have substantially advanced the AMP discovery landscape, by allowing mining of large-scale -omics datasets through the application of predictive models based on comparative analyses with sequences and structures of known AMPs (reviewed by Ramazi et al., 2022).

2.1. Traditional approaches to nematode AMP identification

Traditionally, nematode AMP data have been derived from model species, including the free-living *Caenorhabditis elegans* and the swine roundworm *A. suum* (see Lee et al., 1989; Ewbank and Zugasti, 2011); more recently, broader *in silico* studies in a wide range of nematode species have revealed five distinct AMP groups: (i) Cecropins; (ii) Cysteine-stabilised α-helix and β-sheet Fold (CSαβ) peptides (formerly known as defensins); (iii) Nemapores; (iv) Diapausins, and (v) Glycine Rich Secreted Peptides (GRSPs) (Lee et al., 1989; Kato and Komatsu, 1996; Bányai and Patthy, 1998; Andersson et al., 2003; Couillault et al., 2004; Dieterich et al., 2008; Tarr, 2012a; Ying et al., 2016; Gu et al., 2018; Irvine et al., 2023b). Each AMP group is characterised by key sequence features; for example, CSαβ peptides, Nemapores and Diapausins are defined by cysteine motifs (Tanaka et al., 2003; Tarr, 2012b; 2016; Irvine et al., 2023b), Cecropins are linear amphipathic peptides

with alpha-helical structures (Tarr, 2012b), and GRSPs are linear with an enrichment of glycine residues (Ying et al., 2016). Historically, *in silico* studies identified nematode AMPs using draft genomes and expressed sequence tag (EST) datasets available at the time of analyses (Tarr, 2012a, 2012b, 2016). However, at present, our ability to uncover nematode AMPs is significantly enhanced by the availability of nematode genome and transcriptome datasets (Doyle, 2022). Indeed, the improvement and expansion of nematode *in silico* data have enabled exploration of species-specific and pan-phylum putative AMP-encoding gene diversity (Rončević et al., 2022; Irvine et al., 2023b; Mladineo et al., 2023; Rooney et al., 2024). A recent study revealed that nematodes are AMP-rich, with >5000 putative AMP-encoding genes identified across 109 nematode species; of these, 90 % were novel and representative of all five known AMP groups (Irvine et al., 2023b).

2.2. Novel approaches to nematode AMP identification

Classical *in silico* approaches to AMP identification are progressively being supported by computational prediction strategies. The amalgamation of machine and deep learning algorithms into existing AMP prediction tools and their integration into accessible AMP databases has accelerated AMP identification and enhanced tool capability and reliability. Whilst the Antimicrobial Peptide Database (APD) provided the first opportunity to predict the antimicrobial potential of peptides based on physicochemical properties (*e.g.* charge, hydrophobicity), >10 open-source AMP prediction tools are now available, including: iAMP-2 L (Xiao et al., 2013), AMPA (Torrent et al., 2012), $DBAASP_{v3.0}$ (Pirtskhalava et al., 2021), iAMPpred (Meher et al., 2017), Antimicrobial peptide $scanner_{vr.2}$ (Veltri et al., 2018), AmPEP (Bhadra et al., 2018), AI4AMP (Lin et al., 2021) and sAMPpred-GAT (Yan et al., 2023). While open-source AMP prediction tools allow for rapid identification of putative AMPs from genomic data, they are reliant on the quality of the predicted proteomes used as query sequences. In addition, the accuracy of AMP prediction tools is inextricably linked to the quality of software training datasets (Sidorczuk et al., 2022).

The application of these unique integrated computational pipelines to predicted protein datasets from 96 nematode species has spurred the discovery of novel nematode-derived AMPs (Irvine et al., 2023a). Using this approach, 15,339 putative AMP-like peptides (AMP-LPs), which lacked sequence similarity to known AMP groups, were identified; a subset of these ($n = 16$) were synthesised and tested in antimicrobial assays, of which six exhibited antibacterial activity (MIC<100 μg/ml) against at least one bacterial species, and four (nAMP-LP-18, nAMP-LP-249, nAMP-LP-298, nAMP-LP-617) displayed notably potent (MIC <64 μg/ml) and broad-spectrum antibacterial properties (Irvine et al., 2023a). These data demonstrate the application of integrated computational AMP prediction pipelines to identify novel nematode-derived bioactive AMPs, and provide confidence in this approach for translation to other biological systems.

In silico identification of nematode AMP-LPs has provided a wealth of potential AMP candidates (Irvine et al., 2023a). However, validation of the antimicrobial activities of identified AMP-LPs remains a challenging but necessary step to confirm the biological activity of these peptides. The original study (Irvine et al., 2023a) prioritised AMP-LPs to test in experimental assays based on homology-based curation, Gene Ontology annotation and peptide length. Alternative strategies of AMP-LP prioritisation could include computational prediction of peptide structure (McDonald et al., 2022), and/or evidence of transcriptomic or proteomic expression of AMP-LPs in nematode tissues and/or biofluids.

3. Diversity, classification, and biochemical properties of nematode AMPs

3.1. Cecropins

The first nematode AMP, known as Cecropin P1, was originally isolated from the swine intestinal tract, and reported as the first mammalian Cecropin. Subsequent analyses revealed that Cecropin P1 was derived from *A. suum* (Andersson et al., 2003); however, due to its initial designation as a mammalian peptide, it became one of the most widely studied AMPs.

Bioinformatic analyses in nematodes indicated that the Cecropins are restricted to ascarid species, with homologues initially described in the human and canine roundworms *Ascaris lumbricoides* and *Toxocara canis*, respectively (Pillai et al., 2005; Tarr, 2012b). More recent *in silico* analyses also identified Cecropins from the closely related genera *Anisakis* and *Parascaris* (Rončević et al., 2022; Irvine et al., 2023b). The restriction of Cecropins to the Ascaridoidea suggests Ascarid-specific roles that require further functional characterisation.

Ascaris suum Cecropin P1 displays potent antimicrobial activities against bacteria, fungi and viruses (Cirioni et al., 1998; Giacometti et al., 2000; Guo et al., 2014). Cecropin P1 is primarily α-helical (Table 1) and permeabilizes bacterial cell membranes *via* a 'carpet-like' mechanism, whereby the bacterial membrane surface is covered by the AMP that, upon reaching a critical concentration, disrupts the orientation of phospholipid molecules (Gazit et al., 1995, 1996). Intriguingly, transformation of several economically relevant animals and plants [*e.g.* tobacco (Zakharchenko et al., 2005), potato (Zakharchenko et al., 2007) and rainbow trout (Chiou et al., 2014)] to express Cecropin-encoding genes has resulted in mutant organisms with increased resistance to pathogens (Table 1).

3.2. Diapausins

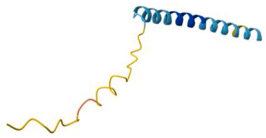
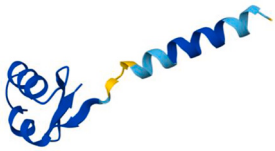
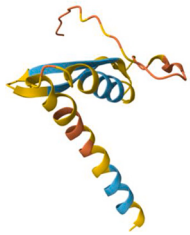
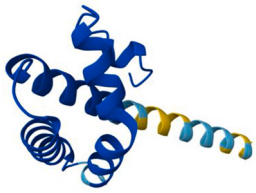
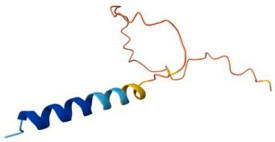
Nematode Diapausins were originally identified from the genome of the free-living nematode *Pristionchus pacificus* (Dieterich et al., 2008). More recently, pan-phylum *in silico* analyses revealed that Diapausins are restricted to soil-dwelling free-living nematodes of Clade 9/V and 11/IV, except *Caenorhabditis* spp. (Irvine et al., 2023b). The absence of Diapausin-encoding genes in *C. elegans* is peculiar, given the similar environmental niche occupied by these species, and highlights the need for further research into the structure and function of AMPs beyond model nematode species.

Diapausin peptides are cysteine-stabilised (Table 1), and homologous to an antifungal diapause-specific peptide from the beetle, *Gastrophysa atrocyanea* (Tanaka et al., 2003). Interestingly, *P. pacificus* shares a necromenic association (= symbiotic relationship in which an organism lives in association with a host which eventually dies) with scarab beetles (Table 1); this observation suggests that *Pristionchus* spp. may have acquired Diapausin-encoding genes *via* horizontal gene transfer from beetles (Rödelsperger and Sommer, 2011). This hypothesis is supported by the occurrence of a putative Diapausin-like sequence in an insect iridovirus (Tanaka et al., 2003). Diapausin-encoding genes have yet to be functionally characterised in nematodes.

3.3. Cysteine-stabilised α-helix and β-sheet fold peptides

Cysteine-stabilised α-helix and β-sheet fold (CSαβ) peptides (formerly known as Defensins) are a diverse group of peptides that share a conserved structure containing an α-helix and two antiparallel β-sheets stabilised by at least three disulphide bonds (Zhu et al., 2005) (Table 1). The first nematode CSαβ peptide identified was the *A. suum* antibacterial factor (ASABF) isolated from the PCF (Kato, 1995; see Table 1). ASABF (later renamed ASABF-α) is a 71aa peptide with eight cysteine residues that form four disulphide bonds. ASABF-α also shares similarity with arthropod Defensins, including the CSαβ core (Kato and Komatsu, 1996). Subsequently, ASABF-type peptides were identified in *C. elegans* (encoded by *abf-1* and *abf-2*; Kato et al., 2002), and in selected species from all other nematode clades (Tarr, 2012b). Notably, *A. suum* encodes five additional ASABF members (ASABF-β, −γ, −δ, −ε and -ζ; Pillai et al., 2003), alongside a novel member, ASABF-6Cys-α, that contains only six cysteine residues, suggesting a rearrangement of disulphide bonding

Table 1
Overview of five distinct groups of nematode antimicrobial peptides (AMPs), and peptide features. Representative members are provided for each group, including *Ascaris suum* Cecropin-P1, *Pristionchus pacificus* Diapausin, *Ascaris suum* ASABF-α, *Caenorhabditis elegans* SPP-3 and NLP-31. The function(s) of each representative AMP and corresponding AlphaFold-generated 3D structural model are provided. Colour scale indicates per-residue confidence scores for structural predictions.

AMP Group	Peptide features	Representative member and predicted structure	Function(s)
Cecropin	Linear peptides with an α-helical amphipathic structure	*Ascaris suum* Cecropin-P1 	Potent antimicrobial activities against bacteria, fungi and viruses; permeabilizes bacterial cell membranes *via* a 'carpet-like' mechanism
Diapausin	Cysteine-stabilised peptide consisting of a triple-stranded β-sheet and α helical structure	*Pristionchus pacificus* Diapausin 	Antifungal activity in beetles; function in nematodes yet to be established
Cysteine-stabilised α-helix and β-sheet Fold (CSαβ)	α-helix and two β sheet structure stabilised by at least four-six cysteines forming three disulphide bonds	*Ascaris suum* ASABF-α 	Antibacterial activity, particularly against Gram-positive bacteria
Nemapore	Cysteine stabilised peptide containing a Saposin-B like fold	*Caenorhabditis elegans* SPP-3 	Antibacterial activity against a wide range of Gram-negative and Gram-positive bacteria; membrane-permeabilizing and pH-dependent pore-forming activity
Glycine Rich Secreted Peptides (GRSPs)	Secreted peptides with an enrichment of glycine residues	*Caenorhabditis elegans* NLP-31 	Antibacterial and antifungal activities, likely by binding microbial DNA to inhibit DNA synthesis
		Model confidence scores: Very high / High / Low / Very low	

(Minaba et al., 2009). Localisation and expression studies point towards a role for nematode CSαβ peptides in digestion and immunity. Indeed, *C. elegans* ABF-2 is localised to the pharynx and associated with digestion of bacteria and prevention of microbial infection (Kato et al., 2002; Pillai et al., 2003) (Table 1). In addition, ASABF-6Cys-α displays tissue-specific upregulation in response to bacterial challenge, which suggests a role in nematode immunity (Minaba et al., 2009).

Additional peptides that share the CSαβ core, but that are distinct from ASABF-type peptides, have been identified in nematodes. The discovery of Macin-like sequences in nematodes (Gerdol et al., 2012), in addition to Cremycin and Mehamycin antifungal peptides in *Caenorhabditis remanei* and the plant parasitic nematode *Meloidogyne hapla* (Zhu and Gao, 2014) and the eight-cysteine CSαβs in the free-living nematode *Panagrolaimus superbus* (Tarr, 2016), further expanded the CSαβ group of nematode AMPs. As a result, Tarr (2016) devised a cysteine reference array to improve delineation between peptides containing the CSαβ fold, thus defining four main subgroups: Mollusc Nematode Defensins (MNDs; including ASABFs), Traditional Insect Defensins (TIDs; including Cremycins), Macins, and Plant/Drosomycin Defensins (Tarr, 2016). A pan-phylum *in silico* analysis later revealed

that, whilst CSαβ-encoding genes are broadly distributed across the Nematoda, with representatives in each of the five clades, individual subgroup profiles are considerably diverse (Irvine et al., 2023b). Beyond the MNDs, limited functional information is available for other CSαβ subgroup members. Traditional insect defensins have only been functionally explored in *Caenorhabditis* spp., with initial studies indicating that *C. remanei* Cremycin-5 displays potent antifungal activity (inhibitory activity at <20 μM) but no antibacterial activity (Zhu and Gao, 2014); nematode Macin and Drosomycin peptides have yet to be functionally characterised.

3.4. Nemapores

Other cysteine-stabilised AMP groups lacking a CSαβ core are present in nematodes. These include homologues of pore forming peptides from the amoeba *Entamoeba histolytica* (Amoebapores) that were initially identified in *C. elegans* (and designated Caenopores; Bányai and Patthy, 1998). Caenopores were subsequently renamed as Nemapores to reflect their occurrence across nematode species. Nemapores belong to the Saposin-like protein (SAPLIP) superfamily, which are characterised by the presence of a Saposin B domain, consisting of six cysteine residues that form three disulphide bonds (Bruhn, 2005) (Table 1). In addition to proteins with antimicrobial activity, the SAPLIP superfamily also includes other functional proteins, such as those responsible for lipid catabolism, surfactant stabilisation and sphingomyelin hydrolysis (Michalek and Leippe, 2015).

A homology-directed study demonstrated that Nemapores were present in 46 of the 58 nematode species examined and identified some proteins with multiple Saposin B domains (Tarr, 2012b). Saposin B domain proteins can be processed to produce individual peptides in the amoeba *Naegleria fowleri* (Herbst et al., 2004), albeit this has not been demonstrated in nematodes. A recent pan-phylum *in silico* analysis conducted by Irvine et al. (2023b) indicated that Nemapores are the most widely distributed AMP group within the phylum Nematoda. Indeed, at least one putative Nemapore-encoding gene was identified from each of the 109 nematode species examined, highlighting a possibly conserved role for Nemapores in nematode immunity. Significantly, functional studies in *C. elegans* have demonstrated that several Nemapores possess antibacterial activity against a range of Gram-positive and -negative bacterial species (Bányai and Patthy, 1998; Roeder et al., 2010; Hoeckendorf and Leippe, 2012). However, functional divergence between individual Nemapores is evident, including roles in development and egg output in *C. elegans* (Roeder et al., 2010; Hoeckendorf and Leippe, 2012) (Table 1).

3.5. Glycine rich secreted peptides

The Glycine Rich Secreted Peptide (GRSP) group in nematodes broadly encompasses genes encoding prepropeptides containing fewer than 200 amino acids, a signal peptide and a glycine rich mature peptide region (Ying et al., 2016) (Table 1). Initially, this group included the YGGW-amide family of the neuropeptide-like proteins (NLPs), and closely related peptides known as Caenacins (Tarr, 2012a). Homology analyses of EST datasets identified putative antimicrobial NLP/Caenacin sequelogues in eight other nematode species, including *Bursaphelenchus xylophilus*, *Anisakis simplex*, *Globodera rostochiensis* and *Strongyloides ratti* (McVeigh et al., 2008).

The GRSP group is highly expanded in *Caenorhabditis* spp. (Ying et al., 2016). While it is challenging to establish phylogenetic relationships between members of the GRSP group, mainly because of short sequence length and highly repetitive glycine repeats, genes encoding GRSPs have been hypothesised to be under positive selection pressure and undergoing rapid evolution (Pujol et al., 2012). Recent homology analyses indicated that putative GRSPs are highly abundant across the Nematoda (>3000 GRSP-encoding genes), with species within Clade 9/V and Clade 10/IV exhibiting the highest number of putative GRSP-encoding genes (Irvine et al., 2023b). Nevertheless, it is unclear whether the putative GRSPs identified share an evolutionary origin, and further analyses are required to resolve their evolutionary history.

The GRSPs have been functionally linked to several aspects of the biology of *C. elegans*. *Caenorhabditis elegans* exposure to the bacterial and fungal pathogens *Serratia marcescens* and *Drechmeria coniospora*, respectively, induced expression of *nlp-29* (Couillault et al., 2004). Silencing of *nlp-31* resulted in increased susceptibility to the Gram-negative pathogen *Burkholderia pseudomallei* (Lim et al., 2016); indeed, synthetic NLP-31 was shown to exert antibacterial and antifungal activities (Couillault et al., 2004), likely by binding microbial DNA to inhibit DNA synthesis. Interestingly, NLP-29 also plays a role in age-associated neuron degeneration (Lezi et al., 2018) and sleep in *C. elegans* (Sinner et al., 2021), indicating that members of the GRSP group may be characterised by diverse functions beyond antimicrobial activity (Table 1).

Whilst research to date indicates that nematodes encode and express an extensive repertoire of AMPs with potential relevance to nematode biology and host-parasite-environment interactions, relatively little is known about their structures and functions. Determining the tertiary structures of nematode AMPs will be key to understanding their antimicrobial function(s) and mechanism(s) of action and, in turn, the potential for therapeutic application and exploitation. Harnessing advances in biotechnology and experimental tools, including in AMP detection, synthesis and production will underpin progress in this area.

4. AMP detection in biologically relevant tissues and biofluids

Through innovations in analytical chemistry such as advanced mass spectrometry (LC-MS/MS), peptidomics has become a central tool for the identification of AMPs and is often integrated with bioinformatics to assist the characterisation of peptide structural and physicochemical properties. The development of novel peptidomics pipelines for nematode biofluids that integrate nematode -omics resources and species-specific peptide libraries with LC-MS/MS technology innovations will aid understanding of nematode biology to drive drug target discovery. The majority of nematode peptidomics analyses have focused on whole-worm lysates (Van Bael et al., 2018; Buzy et al., 2021; Cockx et al., 2023), while nematode biofluids remain understudied. Recent studies have applied LC-MS/MS peptidomics pipelines to the analysis of nematode biofluids, specifically *A. suum* PCF, to detect peptides, including AMPs (Jarecki et al., 2011; Atkinson et al., 2021; Irvine et al., 2023a) (see Table 2).

Advances in LC-MS/MS peptidomics pipelines will facilitate translation to other nematode biofluids, including ESPs, for which comprehensive protein (>10 kDa) profiles already exist (Fig. 1) (Moreno et al., 2011; Soblik et al., 2011; Chehayeb et al., 2014; Sotillo et al., 2014; Midha et al., 2018; Ferguson et al., 2024). Indeed, the development of advanced MS technologies such as trapped ion-mobility time-of-flight

Table 2
Summary of antimicrobial peptides (AMPs) identified in parasitic nematode biofluids currently available in published literature. Experimentally validated AMPs are indicated by an asterisk (*).

Parasitic nematode species	AMPs characterised	Reference
Ascaris suum	35 peptides, 6 of which with 'high-confidence'	Atkinson et al. (2021)
Ascaris suum	14 peptides, including Cecropin and CSαβ AMP groups	Midha et al. (2018)
Ascaris suum	20 peptides, 8 of which with demonstrated antibacterial activity*	Irvine et al. (2023a)
Teladorsagia circumcincta	Histone H2A, destabilase, saposin-b domain containing peptide*, metridin ShK toxin domain-containing peptide*, cysteine-rich protein	Rooney et al. (2024)

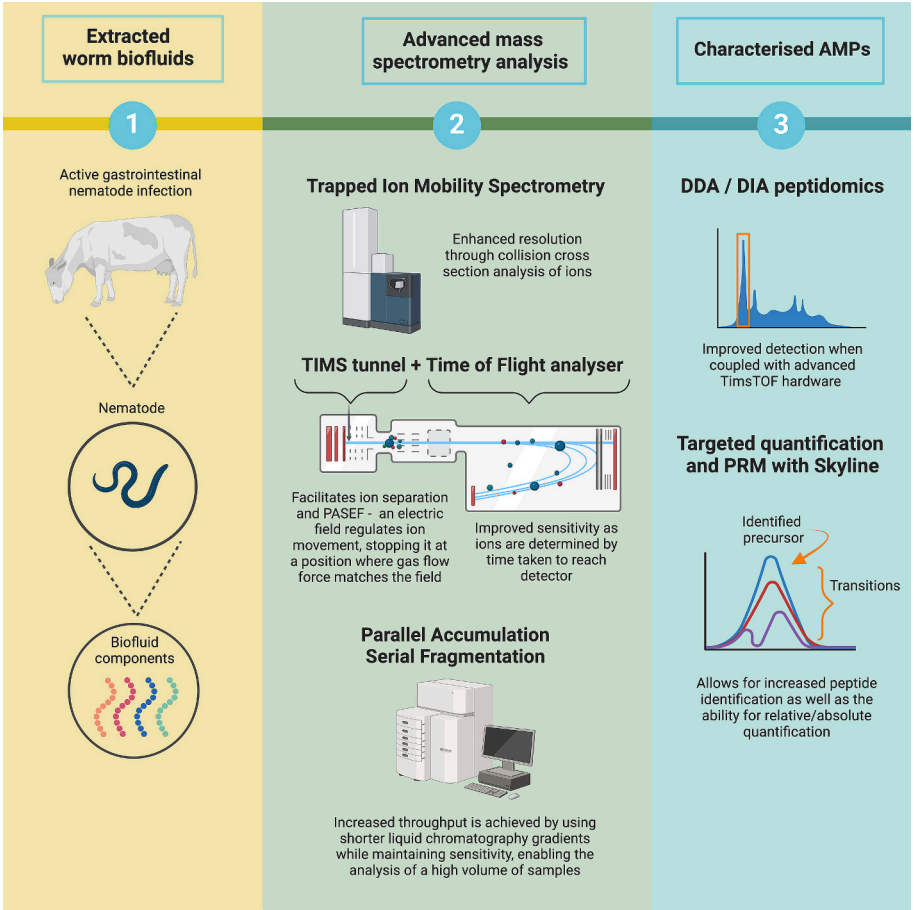


Fig. 1. Outlining advances in mass spectrometry for the characterisation of antimicrobial peptides (AMPs) extracted from worm biofluids. A proposed workflow for analysing antimicrobial peptides (AMPs) from nematode biofluids using advanced mass spectrometry. **(1) Extracted worm biofluid sample**: Biofluids containing AMPs are collected from nematodes infecting host organisms. **(2) Advanced mass spectrometry analysis**: Trapped Ion Mobility Spectrometry (TIMS) enhances resolution by separating ions based on their collision cross-section. The TIMS tunnel combined with Time of Flight (TOF) analysis measures ion mass-to-charge ratios with improved sensitivity. Parallel Accumulation Serial Fragmentation (PASEF) enables higher throughput through shorter liquid chromatography gradients. **(3) Characterised AMPs**: Data-Dependent Acquisition (DDA) and Data-Independent Acquisition (DIA) peptidomics pipelines enhance peptide detection, while targeted quantification with Parallel Reaction Monitoring (PRM) using Skyline software allows for relative and absolute quantification of AMPs. Created in BioRender. https://BioRender.com/i22j226.

(TimsTOF), in combination with novel parallel serial fragmentation (PASEF), provides an additional layer of separation, increases sensitivity, and maximises ion beam capability (Meier et al., 2018, 2020) allowing for improved detection of AMPs in nematode biofluids. Indeed, TimsTOF instruments have been key to immunopeptidomics analyses when compared to traditional orbitrap instruments (Adams et al., 2024). To date, nematode biofluid peptidomics analyses have applied a Data Dependent Acquisition approach (DDA) using a topN method, where only the most abundant precursors are selected for fragmentation. However, the inherent semi-stochastic nature of DDA may lead to lack of reproducibility and make the detection of low abundance peptides challenging, as precursor selection may vary across runs (Tabb et al., 2010). Development of Data Independent Acquisition (DIA) approaches, where every precursor ion is fragmented regardless of its abundance and/or intensity, holds the potential to overcome these limitations (Gillet et al., 2012; Ludwig et al., 2018). Indeed, compared to DDA, DIA is generally characterised by increased reproducibility (Bruderer et al., 2015) and sensitivity (Gillet et al., 2012). Nevertheless, since DIA generates large volumes of data, downstream analyses may prove complex. Future studies may investigate the application of DIA approaches to isolate and profile low-abundance AMPs within nematode biofluids.

Targeted MS methods, such as parallel reaction monitoring (PRM), have recently been applied to nematode neuropeptidomics, yielding a 2–3.5-fold increase in peptide detection compared to traditional non-targeted approaches such as DDA (Cockx et al., 2023). PRM is yet to be applied to the discovery of nematode AMPs and may provide an avenue for both relative and absolute quantification of AMPs in nematode biofluids in response to dynamic biotic and abiotic factors within the host environment.

5. AMP synthesis and production

The synthesis and production of AMPs involve a range of methodologies, each tailored to the specific structural and biochemical characteristics of the target peptide. Synthetic methods offer precision and scalability for linear peptides, but they are limited in handling complex modifications, such as disulphide bonds. Conversely, recombinant techniques leverage cellular systems—prokaryotic (*e.g. E. coli*) or eukaryotic (*e.g.* yeast, insect, and mammalian cells)—to produce AMPs with more intricate structures, albeit with higher costs and additional challenges, such as endotoxin contamination or cytotoxic effects of the AMPs on host cells. Regardless of the production method, purification remains a critical step, requiring tailored protocols to ensure high purity and structural integrity of the peptide for downstream functional applications.

5.1. Synthetic production of AMPs

Synthetic production of AMPs may be achieved *via* liquid-phase synthesis, solid-phase synthesis, and heterologous recombinant expression; each of these techniques presents *pros* and *cons*, and the choice of the most suitable technique often depends on the structural and biochemical characteristics of the target AMP(s). The average length of AMPs contained in the APD is ~30 amino acid residues, and AMPs of ≤ 30 amino acids in length are considered short (Wang et al., 2016; Hussain, 2022). Liquid-phase synthesis is suitable for production of short-chain peptides due to lower cost and the ability to achieve high purity profiles (Bayer and Mutter, 1972; Carratalá et al., 2020). However, this technique involves longer cycle times, several operational steps, and often results in low yields of synthetic peptide (Tymecka and Misicka, 2020). Conversely, solid-phase synthesis is often applied to large-scale production of short- to medium-length AMPs. This approach allows for the precise production of peptides according to their chemical structure and, as it does not require the isolation of intermediates, can be fully automated and scaled up efficiently (Münzker et al., 2017). Overall, directly synthesised peptides meet the desired specifications without the need for extensive downstream processing to overcome challenges linked to production and purification. However, production by synthesis is mostly limited to generating linear short-chain peptides lacking disulphide bonds or other post-translational modifications (PTMs), as most PTMs which occur in complex peptides cannot be carried out (Boto et al., 2018).

5.2. Recombinant expression of AMPs

Whilst peptides ≤ 30 amino acids in length are generally more efficiently produced through chemical synthesis, longer peptides and proteins are more effectively manufactured using recombinant expression techniques (Hoelscher et al., 2022). Such techniques may achieve high purity levels of complex peptides and proteins; however, they are relatively costly and labour-intensive (Münzker et al., 2017). Furthermore, the selection of an appropriate vector for expression is highly influenced by AMP characteristics such as structure, abundance of disulphide bonds, distribution of acidic and basic residues, and isoelectric point (Oliveira et al., 2020). Of note, contamination of recombinant proteins with therapeutic purposes by endotoxins may be of concern (Bonhomme et al., 2024), as even minimal levels of contamination can induce aberrant responses, and result in a range of deleterious effects across cell lines and experimental subjects. Furthermore, the selection of suitable expression vectors depends on scale of production, necessary PTMs, cost considerations, and intrinsic advantages and limitations of each system (Table 3).

Several expression systems are available for recombinant peptide and protein production; however, heterologous expression of AMPs is commonly conducted using eukaryotic and prokaryotic systems such as *E. coli*, *Bacillus subtilis*, and *Komagataella phaffii* (syn. *Pichia pastoris*) (reviewed by Gomes et al., 2016). Of these, *E. coli* has been applied to the synthesis of >80 % of all recombinant AMPs expressed over recent decades (Zhang et al., 2021). Nevertheless, bacterial expression is often unsuitable for production of small peptides at high concentrations, due to the potential toxicity of AMPs to the host cells and the inability to carry out most PTMs (Table 3) (Li, 2011).

Yeast systems, including *Saccharomyces cerevisiae* and *K. phaffii*, offer the advantage of eukaryotic modifications as well as a relatively low cost of production (Table 3). Alternatively, insect cell hosts, leveraging the baculovirus expression vector system (BEVS), have been pivotal in producing complex eukaryotic peptides and proteins with PTMs akin to mammalian cells (Müller et al., 2015). Mammalian cell systems, particularly Chinese hamster ovary (CHO) cells, are most suitable for production of therapeutic proteins and monoclonal antibodies due to their human-like modifications. Plant expression systems, using both transient and stable transformations, have recently emerged as cost-effective and scalable alternatives for producing recombinant proteins with alleviated cytotoxic effects (Hoelscher et al., 2022). Lastly, cell-free expression systems, utilise extracts from organisms such as *E. coli* and wheat germ, and offer rapid and versatile protein production without the need for living cells (El-Baky et al., 2020).

5.3. Purification of recombinantly expressed AMPs

A significant bottleneck in the production of AMPs *via* recombinant technology is represented by the extraction of high-purity target peptides from a complex mixture of biomolecules native to the expression vector (Tang et al., 2024). Thus, the application of protocols leveraging multiple techniques is necessary to attain the desired purity and quantity of peptide for downstream structural and functional characterisation.

Table 3
Comparative overview of the key recombinant expression systems available for production of antimicrobial peptides (AMPs), including bacterial, yeast, insect, mammalian, plant-based, and cell-free systems. Each vector is evaluated based on its suitability for specific production needs, such as yield, scalability, cost, and capacity for post-translational modifications (PTMs).

Recombinant expression system	Advantages	Limitations	Applications
Bacterial (*E. coli* and *B. subtilis*) (Deng et al., 2017)	Highest yield, lowest cost of production, scalability, potential and rapid incorporation of foreign DNA	Limited potential for PTM, glycosylation or disulphide bond formation; high endotoxin contamination risk, time consuming	Large-scale production of peptides without complex modifications
Yeast [*S. cerevisiae* and *K. phaffii (P. pastoris)*] (Baghban et al., 2019)	High-level expression, complex PTM possible, cost-effective, relatively low cost, low risk of contamination and scalable compared with mammalian cells	Differing glycosylation patterns from higher eukaryotes with possible hyperglycosylation and complex culture conditions with slow production	Production of peptides with eukaryotic modifications
Insect (Baculovirus Expression Vector Systems) (Müller et al., 2015)	Suitable for complex PTM, high-level expression, suitable for difficult-to-express proteins	Higher cost, and longer production times leading to reduced scalability	Production of peptides requiring complex modifications
Mammalian (Chinese Hamster Ovary Cells) (Kim et al., 2012)	Human-like PTM possible, suitable for therapeutic applications	High cost and lower yield compared to bacterial and yeast systems, with complex culture conditions and high contamination risk	Therapeutic peptides requiring human-like modifications
Plant (Plastids) (Hoelscher et al., 2022; Wilbers et al., 2017)	Cost-effective for large-scale production, suitable for complex PTM and low risk of contamination, absence of introgression risk	Variable expression levels, and different glycosylation patterns (but amenable to glycoengineering)	Vaccines and therapeutic peptides requiring modifications
Cell-Free Systems (El-Baky et al., 2020)	Rapid protein production, suitable for toxic proteins, no cell culture maintenance needed	High cost of reagents, limited scalability	Small-scale production of toxic or unstable peptides, and rapid testing and prototyping of peptide production

For instance, the addition of tags to assist affinity-based purification must not interfere with the biological activity of the recombinant protein; in addition, purification workflows must be undertaken considering the physicochemical properties of target peptide(s). For example, the molecular weight and isoelectric point of the target peptide are key to the selection of a suitable size-exclusion chromatography medium and/or ion-exchange chromatography matrix, respectively (Singh and Bose, 2022).

The presence of cysteines associated with the target peptide, whether free or within inter- or intramolecular disulphide bonds, must be determined from the amino acid sequence and 3D structure; reducing agents, such as dithiothreitol must be applied to prevent artefactual disulphide bond formation in the presence of free cysteines, but avoided in the presence of disulphide bonds to prevent conformational changes in the target peptide or protein (Remans et al., 2022). Relative peptide stability must also be considered when selecting the appropriate pH, salt content, and temperature conditions during purification and, later, storage (Wingfield, 2015). Whilst several of these variables are determined experimentally, bioinformatics predictive tools based on the amino acid sequences of the target peptide or protein can be of assistance. For instance, ProtParam (https://web.expasy.org/protparam/) estimates the *in vivo* half-life and instability index based on the N-terminal residue of recombinant peptides expressed in systems such as mammalian cells, yeast, and *E. coli*. In addition, the prediction of grand average of hydrophobicity (GRAVY) value is particularly relevant, as reversed phase chromatography (RPC) or hydrophobic interaction chromatography (HIC) can be incorporated within the purification schematic to remove contaminants from highly hydrophobic target peptides.

Recombinant peptide purification is routinely performed by ion exchange (IEX), gel filtration (GF), and ligand affinity (AF) chromatography (Sánchez-Trasviña et al., 2021). These techniques separate peptides based on charge (IEX), size (GF), and ligand affinity (AF) characteristics, respectively. The order in which purification techniques are combined determines the outcome of the separation process and thus should be optimised to the target. However, generally, if *E. coli* is used as the expression vector, a two or three step protocol may be applied post bacterial cell lysis, *e.g.* AF chromatography wherein the selection of a suitable tag is based on the underlying physicochemical properties of the target (commonly used tags include His_6, a tandem fusion tag which binds to the affinity matrix comprising metal ions, GST, or MBP, both of which are larger solubility tags; Pina et al., 2014), followed by IEX chromatography, which separates cations and anions, and separation by hydrodynamic volume or size (size or molecular weight exclusion) by GF chromatography (Sun et al., 2004), or *vice versa*. Although obtaining >95 % pure recombinant peptide for structural and functional studies is a condition usually met through the methods described above, surface adsorption chromatography techniques that exploit the surface hydrophobicity interactions of peptides, such as RPC and HIC, might be applied in combination with other purification strategies for highly hydrophobic peptides that are otherwise challenging to purify and characterise (Singh and Bose, 2022).

6. AMP bioactivity, mechanism of action, and biological function

The antimicrobial actions of AMPs can be broadly divided into membranolytic and non-membranolytic mechanisms of action. The majority of characterised AMPs act by permeabilising microbial cell membranes. The initial interactions between AMPs and microbial membranes are facilitated by the cationic and amphipathic properties of AMPs such that they are attracted and can bind to anionic cell membranes through electrostatic and hydrophobic forces (Matsuzaki, 2019). Once bound, AMPs modify the membrane to cause permeabilisation. Various models have been suggested to explain AMP-membrane interactions including the 'carpet', 'barrel-stave' and 'toroidal pore' models (Matsuzaki, 2019). Whilst it is plausible that GI nematodes may exert antimicrobial activity, either directly or indirectly, *via* the secretion of AMPs, limited functional data are available on nematode AMP bioactivity, and their mode of action remains largely unknown (Pillai et al., 2005; Roeder et al., 2010; Hoeckendorf and Leippe, 2012; Lim et al., 2016; Rooney et al., 2024). Selected techniques for experimental validation of putative AMP leads are reviewed here, including optical density measurements (OD_{600}), colony forming unit (CFU) assays, time-lapse microfluidics, and functional genomics.

6.1. Bioactivity and mechanism of action of AMPs

Since most AMPs are cationic and favour interactions with anionic bacterial membranes, their mechanism of action is thought to consist in the permeabilization of the cytoplasmic membrane (CM), thus resulting in growth inhibition and cell death. These interactions depend on the phospholipid composition of bacterial membranes that can be reconstituted in empirically predictable models. Biophysical studies of AMP interactions with synthetic lipid bilayers have provided insights into the mechanisms of bilayer disruption. Giant unilamellar vesicles (GUVs, with diameters ranging between 1 and 100 μm) are particularly useful to elucidate and confirm the modes of action of AMPs (Moghal et al., 2020). For rapid and high-throughput analysis of AMP mode of action on lipid bilayer models, integrated lab-on-a-chip multilayer microfluidic platforms have been developed, that allow quantification of the membranolytic efficacy of these molecules (Al Nahas et al., 2019; Al Nahas and Keyser, 2021). While synthetic lipid bilayer models and high-throughput analysis of AMP mode of action using microfluidic approaches have enabled user-friendly reliable and fast operation, there are significant disparities between AMP permeabilization of lipid vesicles and killing mechanisms of whole bacterial cells (Faust et al., 2017).

Most mechanistic studies of the bacteriostatic and bactericidal effects of AMPs on whole bacterial cells have focused on bulk, planktonic cultures. Optical density (OD_{600}) measurements are a commonly used technique, where growth inhibition of microbial cultures upon exposure to a given antimicrobial agent is monitored spectrophotometrically, thus offering a rapid and high-throughput assessment of antimicrobial activity (Li et al., 2020). However, since unviable (*e.g.* dead or deformed) bacterial cells may also be included in the final readings, false negative results are possible. Moreover, the OD-based method often lacks sensitivity during the early stages of bacterial growth, which crucially corresponds with the phase of maximum AMP activity (Implen, 2020). Thus, CFU assays, which are used to quantify the number of viable bacteria, may provide a more reliable measure of antimicrobial potency (da Silva et al., 2013). Indeed, one study utilised CFU assays to carry out functional analyses of newly characterised nematode-derived AMPs, that resulted in significant growth reductions of *E. coli* upon exposure to recombinantly expressed AMPs from *T. circumcincta* and *H. contortus*, respectively (Rooney et al., 2024). Nevertheless, it is worth noting that, since manual colony counting is prone to human error, the use of automatic colony counters may assist in reducing possible miscalculations.

While bulk assays provide valuable insights into various biophysical and biochemical events, they are characterised by significant limitations. These include poor time resolution, lack of single-cell analysis, absence of spatial information, and susceptibility to inoculum effects, which often results in an averaged view that obscures the dynamics of AMP action. For instance, while some AMP-induced damage occurs within seconds (Fantner et al., 2010), bulk assays typically resolve events over minutes, thus missing critical early dynamics of AMP-cell interactions. Moreover, bulk assays fail to account for cell-to-cell heterogeneity, a crucial factor in determining AMP efficacy, and provide no subcellular spatial information that is key to understanding AMP mechanism(s) of action.

Time-resolved single-cell imaging may instead offer dynamic and spatial insights into AMP interactions with live bacterial cells. These

assays use time lapse microscopy to observe the effects of AMP exposure upon a field of well-isolated cells, thus revealing the exact sequence and distribution of events. Label-free imaging modes, such as phase contrast or brightfield imaging, can be applied to track the growth and morphology of the target individual bacterial cells, and thus analyse the growth-halting and/or lytic effects of AMPs (Chileveru et al., 2015). Additionally, multicolour fluorescence imaging can be used to monitor different compartments and components of bacteria (Sochacki et al., 2011; Rangarajan et al., 2013) and fluorescently tagged AMPs, thus enabling direct studies of their binding and binding-induced perforation locations (Kandaswamy et al., 2013; Scheinpflug et al., 2015). This approach has allowed a detailed analyses of the physiological response of both Gram-positive and -negative bacterial cells towards AMPs (Barns and Weisshaar, 2013; Choi et al., 2015, 2017; Yang et al., 2018; Zhu et al., 2019), thus highlighting the necessity of studying physiological dynamics of whole individual bacterial cells beyond bulk and model bilayer studies.

High-throughput single-cell microfluidic systems have also emerged as powerful tools for functional characterisation of AMPs, robust statistical analyses of AMP action kinetics, and assessment of underlying spectrum of physiological heterogeneity. These systems allow for the precise control of microenvironmental conditions, the ability to maintain large numbers of cells under spatially uniform growth conditions and rapid timelapse scanning of multiple fields of view to collect high throughput data of AMP action dynamics. Amongst currently available microfluidic platforms, the 'mother machine' houses individual cell lineages in hundreds of isolated chambers (Wang et al., 2010). This platform has provided highly sensitive mechanistic analysis of AMP action (Cama et al., 2022). Recently, single-cell analysis of nematode AMP action using the mother machine was critical to distinguish possible underlying AMP mode-of-action against Gram-positive strains (Rooney et al., 2024).

Nevertheless, one limitation of microfluidic single-cell time lapse studies is represented by the focus on planktonic growth. Indeed, microbes inhabiting the mammalian gut often grow under anaerobic conditions, transition through feast-famine cycles, or remain dormant within multi-species, biofilm-like environments. Therefore, microfluidic platforms that simulate such events are likely to reveal additional, context-dependent insights into AMP modes of action. For example, a study by Choi et al. (2015) developed an anaerobic chamber to simulate the anaerobic growth conditions of bacterial cells, revealing insights into the role of oxidative stress in AMP-mediated cell death. A recent study further adapted the mother machine platform to simulate the complex growth conditions of feast-famine cycles (Bakshi et al., 2021), thus enabling detailed analysis of antibiotic action during dormant phases *vs.* feast-famine cycles. These studies provide direct links between cell physiology and physiological heterogeneity and susceptibility to AMPs. Additionally, two-dimensional microcolonies provide insights into cell-cell interactions and spatial organisation, both of which are likely to influence AMP action *in vivo* (Binder et al., 2017; Cooper et al., 2018). Microfluidic designs that closely simulate the complex, polymicrobial nature of biofilms will further elucidate the roles of microbiome composition and biofilm structure in determining AMP mode of action (Wright et al., 2015; Ugolini et al., 2024).

The antibacterial activity of AMPs should be assessed against several strains of Gram-positive and -negative bacteria; indeed, previous investigations have shown that *A. suum* and *H. polygyrus* secreted AMPs exert varying degrees of antibacterial activity against a range of microbes (Midha et al., 2018; Rausch et al., 2018). Furthermore, comprehensive functional analyses of AMP activity should be carried out on panels of bacterial, protist, viral, and fungal microorganisms, including pathogens (*e.g.* Klubthawee et al., 2020; Balmeh et al., 2021; Ul Haq et al., 2024). Additional functional characterisation experiments may provide insights into the antimicrobial properties of nematode AMPs. Notably, Midha et al. (2018) demonstrated the antimicrobial and antibiofilm activity of *A. suum* secreted AMPs and antimicrobial proteins (*i.e.* Cecropins, ASABFs, C-type lectins and galactins) using radial diffusion assays, and assessed the impact of nematode AMPs on biofilm formation using a crystal violet assay; crystal violet is used to stain bacterial cells and bacterial extracellular components following exposure to the treatment (*e.g.* AMP) or PBS control, where differences in stain intensity are indicative of the degree of biofilm disruption and/or inhibition of biofilm formation induced by AMPs (Xu et al., 2021). Similarly, the ability of AMPs to inhibit the growth of bacteria may be evaluated using agglutination assays; in these assays, bacteria are exposed to AMPs in the presence and absence of $CaCl_2$ and their subsequent aggregation is assessed using microscopy (Midha et al., 2018). Moreover, the effect of treatment with nematode AMPs on the membrane integrity, morphology and viability of bacteria could be further assessed using transmission electron microscopy (TEM), to provide further insights into the mechanism of action (Wang et al., 2022b).

6.2. The role of AMPs in nematode biology

Understanding the function of nematode AMPs is critical to their exploitation as potential novel therapeutics for the treatment of pathogenic microorganisms and/or the control of nematode parasites. Indeed, the reliance of nematode parasites on innate immunity offers an opportunity to exploit the perceived importance of AMPs in this context as novel drug targets. The application of advanced reverse genetics tools (*e.g.* RNAi, CRISPR) to nematode parasites offers a route to explore the function of AMPs in nematode biology, infection, and interactions with the vertebrate host.

Several studies have exploited reverse genetics to explore nematode AMP function. A number of *C. elegans* AMP-encoding genes have been silenced *via* RNAi including the Caenopores (*spp-5* and *spp-3*) revealing links to nutrition and egg laying (Roeder et al., 2010; Hoeckendorf and Leippe, 2012), and a Neuropeptide Like Protein (NLP-31) where the mutants displayed increased susceptibility to bacterial pathogens (Lim et al., 2016). No other studies have reported nematode AMP-encoding gene function using reverse genetics approaches. Indeed, a significant impediment to functional characterisation of nematode parasite genes is the absence of robust functional genomics tools for the majority of species. Furthermore, most nematode parasites cannot be cultured *in vitro* and are not amenable to reverse genetics tools including RNAi or CRISPR, limiting the scope for functional analyses (Castelletto et al., 2020).

Despite the availability of a validated RNAi platform for *A. suum* (McCoy et al., 2015), AMP silencing has been challenging. Indeed, *A. suum* Cecropin-encoding gene RNAi was inconsistent, possibly due to hypervariability in *A. suum* Cecropin-encoding gene expression (unpublished data, A. Irvine). Efforts to improve parasite culture and functional genomics platforms will enable robust interrogation of parasitic nematode AMP function. *Strongyloides* species have emerged as a tractable parasite model capable of gene knockdown *via* RNAi and gene knockout *via* CRISPR (Castelletto et al., 2020). *Strongyloides* also boast an impressive portfolio of phenotype bioassays that can be harnessed to explore gene function (Cadd et al., 2022). *Strongyloides stercoralis* infection is associated with microbiome changes in human hosts (Jenkins et al., 2018), therefore exploitation of *Strongyloides* reverse genetics may provide an opportunity to explore AMP function in a parasitic nematode *in vivo*. Indeed, the *Strongyloides* genus possesses putative Nemapore-encoding and GRSP-encoding genes as well as >50 AMP-LPs per species (Irvine et al., 2023a, 2023b).

A major limitation in characterising parasite-host-microbiome interactions, including the role of AMPs, is represented by the limited availability of physiologically-relevant models. The recent development of host organoids (physiologically functional, tissue-specific *in vitro* culture systems) and their growing application to helminthology is beginning to address this. Indeed, GI organoids have been developed for several of the major hosts affected by nematode parasites, including human, porcine, bovine, and ovine (Zachos et al., 2016; Pleguezuelos-

Manzano et al., 2020; Vermeire et al., 2021; Faber et al., 2022; Smith et al., 2022). Moreover, these organoids have been adapted to support the growth of microbe co-cultures (Karve et al., 2017; Smith et al., 2021; Co et al., 2021; Nash et al., 2021; Vermeire et al., 2021; Blake et al., 2022; Lacroix-Lamandé et al., 2023) and efforts continue to be directed towards the development of an organoid model that can support microbial communities representative of a tissue-specific microbiome (Fofanova et al., 2024). Organoids are already being exploited to assess the effects of probiotics on resident gut microbes and these approaches could be translated to examine the antimicrobial effects of nematode AMPs (Stewart, 2023). These systems provide the ability to directly capture microbial growth in the host gut and yield quantitative data on virulence and pathogenicity (*e.g.* inflammatory response). It has also been demonstrated that organoids, derived from tissues from a range of parasitic nematode hosts, can sustain active nematode infections, and that the transcriptional profiles of organoid tissues post-nematode infection and/or exposure to worm ESPs reflect those observed *in vivo* (Duque-Correa et al., 2020, 2022; Drurey et al., 2021; Smith et al., 2021; Faber et al., 2022; Karo-Atar et al., 2022). Collectively, development and optimisation of host-nematode-microbe organoids present unique opportunities to model the effect of helminth AMPs on the gut holobiont. Moreover, advanced organoid systems paired with genetically manipulated nematodes hold the potential to revolutionise the understanding of AMP-specific roles in nematode-host-microbiome interactions.

7. Concluding remarks & future directions

Recent progress in *in silico* analyses and computational tools, alongside biotechnological advances in mass spectrometry, peptidomics, single-cell microfluidics, organoid technologies, and reverse genetics have enabled the identification and characterisation of nematode biofluid-derived AMPs to drive understanding of the role(s) of these molecules in host-parasite-microbiome interactions, and seed the development of novel therapeutics for nematode and bacterial pathogens (Fig. 2). The scalability and low-cost of AMP production, particularly of linear short-chain peptide variants, further enhances the feasibility of exploring the potential of AMPs in research and therapeutic applications. In addition, exploitation of novel experimental approaches, including microfluidic single-cell studies and organoid models, may provide new insights into AMP dynamics within complex

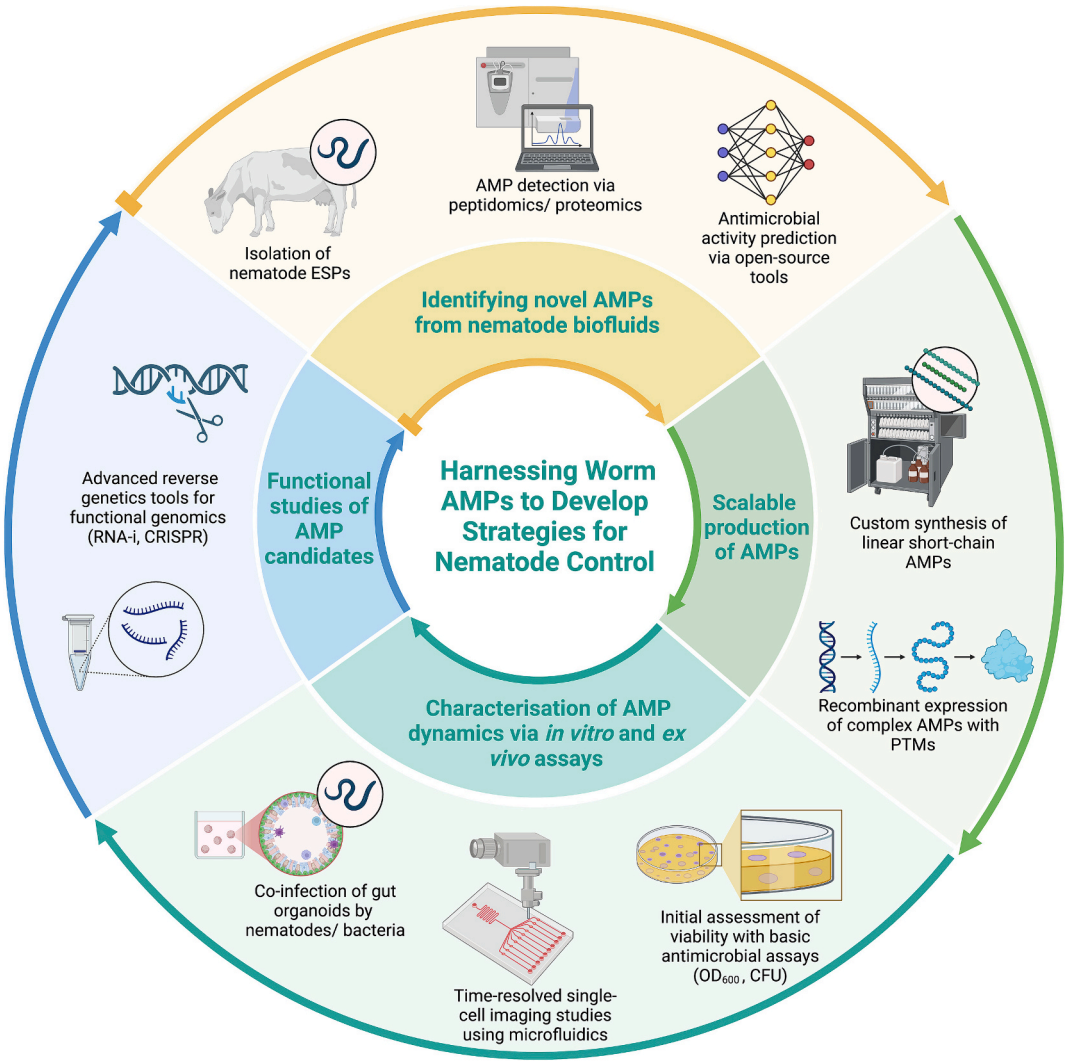


Fig. 2. Harnessing worm antimicrobial peptides (AMPs) to develop strategies for nematode control. Integrated approach to developing antimicrobial peptide (AMP) candidates for nematode control. Novel AMPs can be identified through biochemical isolation, peptidomics analyses, and peptide prediction using bioinformatics. Scalable production can be achieved *via* custom synthesis for short-chain peptides and recombinant expression for complex AMPs with post-translational modifications (PTMs). AMP dynamics are assessed through viability assays, single-cell microfluidics, and organoid models co-infected with nematodes and bacteria. Functional studies of AMP candidates utilise reverse genetics tools, such as RNAi and CRISPR, to understand their roles in nematode biology and applications in control strategies. Created in BioRender. https://BioRender.com/i22j226.

host-parasite-microbiome systems, and may support the development of novel, sustainable therapeutics for nematode control. Furthermore, nematode AMPs may represent a yet untapped source of novel antimicrobials, whose activity against pathogenic microorganisms (including multidrug-resistant bacteria, archaea, fungi, and protozoa) is worth exploring in an effort to improve global human and animal health.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

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References

Adams, C., Gabriel, W., Laukens, K., Picciani, M., Wilhelm, M., Bittremieux, W., Boonen, K., 2024. Fragment ion intensity prediction improves the identification rate of non-tryptic peptides in timsTOF. Nat. Commun. 15 (1), 3956. https://doi.org/10.1038/s41467-024-48322-0.

Al Nahas, K., Keyser, U.F., 2021. Standardizing characterization of membrane active peptides with microfluidics. Biomicrofluidics 15 (4), 041301. https://doi.org/10.1063/5.0048906.

Al Nahas, K.A., Cama, J., Schaich, M., Hammond, K., Deshpande, S., Dekker, C., Ryadnov, G., Keyser, U.F., 2019. A microfluidic platform for the characterisation of membrane active antimicrobials. Lab Chip 19 (5), 837–844. https://doi.org/10.1039/C8LC00932E.

Andersson, M., Boman, A., Boman, H.G., 2003. *Ascaris* nematodes from pig and human make three anti-bacterial peptides: isolation of cecropin P1 and two ASABF peptides. Cell. Mol. Life Sci. 60 (3), 599–606. https://doi.org/10.1007/s000180300051.

Atkinson, L.E., Liu, Y., McKay, F., Vandewyer, E., Viau, C., Irvine, A., Rosa, B.A., Li, Z., Liang, Q., Marks, N.J., Maule, A.G., Mitreva, M., Beets, I., Li, L., Mousley, A., 2021. *Ascaris suum* informs extrasynaptic volume transmission in nematodes. ACS Chem. Neurosci. 12 (17), 3176–3188. https://doi.org/10.1021/acschemneuro.1c00281.

Baghban, R., Farajnia, S., Rajabibazl, M., Ghasemi, Y., Mafi, A., Hoseinpoor, R., Rahbarnia, L., Aria, M., 2019. Yeast Expression Systems: Overview and Recent Advances. Mol. Biotechnol. 61 (5), 365–384. https://doi.org/10.1007/s12033-019-00164-8.

Bakshi, S., Leoncini, E., Baker, C., Cañas-Duarte, S.J., Okumus, B., Paulsson, J., 2021. Tracking bacterial lineages in complex and dynamic environments with applications for growth control and persistence. Nat. Microbiol. 6 (6), 783–791. https://doi.org/10.1038/s41564-021-00900-4.

Balmeh, N., Mahmoudi, S., Fard, N.A., 2021. Manipulated bio antimicrobial peptides from probiotic bacteria as proposed drugs for COVID-19 disease. Inf. Med. Unlocked 23, 100515. https://doi.org/10.1016/j.imu.2021.100515.

Bányai, L., Patthy, L., 1998. Amoebapore homologs of *Caenorhabditis elegans*. Biochim. Biophys. Acta 1429 (1), 259–264. https://doi.org/10.1016/s0167-4838(98)00237-4.

Barns, K.J., Weisshaar, J.C., 2013. Real-time attack of LL-37 on single *Bacillus subtilis* cells. Biochim. Biophys. Acta Biomembr. 1828 (6), 1511–1520. https://doi.org/10.1016/j.bbamem.2013.02.011.

Bayer, E., Mutter, M., 1972. Liquid phase synthesis of peptides. Nature 237 (5357), 512–513. https://doi.org/10.1038/237512a0.

Bhadra, P., Yan, J., Li, J., Fong, S., Siu, S.W.I., 2018. AmPEP: sequence-based prediction of antimicrobial peptides using distribution patterns of amino acid properties and random forest. Sci. Rep. 8 (1), 1697. https://doi.org/10.1038/s41598-018-19752-w.

Binder, D., Drepper, T., Jaeger, K.-E., Delvigne, F., Wiechert, W., Kohlheyer, D., Grünberger, A., 2017. Homogenizing bacterial cell factories: analysis and engineering of phenotypic heterogeneity. Metab. Eng. 42, 145–156. https://doi.org/10.1016/j.ymben.2017.06.009.

Blake, R., Jensen, K., Mabbott, N., Hope, J., Stevens, J., 2022. The development of 3D bovine intestinal organoid derived models to investigate *Mycobacterium avium* ssp *paratuberculosis* pathogenesis. Front. Vet. Sci. 9, 921160. https://doi.org/10.3389/fvets.2022.921160.

Bonhomme, D., Cavaillon, J.-M., Werts, C., 2024. The dangerous liaisons in innate immunity involving recombinant proteins and endotoxins: examples from the literature and the *Leptospira* field. J. Biol. Chem. 300 (1), 105506. https://doi.org/10.1016/j.jbc.2023.105506.

Boto, A., De La Lastra, J.M.P., González, C.C., 2018. The road from host-defense peptides to a new generation of antimicrobial drugs. Molecules 23 (2), 311. https://doi.org/10.3390/molecules23020311.

Bruderer, R., Bernhardt, O.M., Gandhi, T., Miladinović, S.M., Cheng, L.-Y., Messner, S., Ehrenberger, T., Zanotelli, V., Butscheid, Y., Escher, C., Vitek, O., Rinner, O., Reiter, L., 2015. Extending the limits of quantitative proteome profiling with data-independent acquisition and application to acetaminophen-treated three-dimensional liver microtissues. Mol. Cell. Proteomics 14 (5), 1400–1410. https://doi.org/10.1074/mcp.M114.044305.

Bruhn, H., 2005. A short guided tour through functional and structural features of saposin-like proteins. Biochem. J. 389 (Pt 2), 249–257. https://doi.org/10.1042/BJ20050051.

Buzy, A., Allain, C., Harrington, J., Lesuisse, D., Mikol, V., Bruhn, D.F., Maule, A.G., Guillemot, J.-C., 2021. Peptidomics of *Haemonchus contortus*. ACS Omega 6 (15), 10288–10305. https://doi.org/10.1021/acsomega.1c00650.

Cadd, L.C., Crooks, B., Marks, N.J., Maule, A.G., Mousley, A., Atkinson, L.E., 2022. The *Strongyloides* bioassay toolbox: A unique opportunity to accelerate functional biology for nematode parasites. Mol. Biochem. Parasitol. 252, 111526. https://doi.org/10.1016/j.molbiopara.2022.111526.

Cama, J., Al Nahas, K., Fletcher, M., Hammond, K., Ryadnov, M.G., Keyser, U.F., Pagliara, S., 2022. An ultrasensitive microfluidic approach reveals correlations between the physico-chemical and biological activity of experimental peptide antibiotics. Sci. Rep. 12 (1), 4005. https://doi.org/10.1038/s41598-022-07973-z.

Carratalá, J.V., Serna, N., Villaverde, A., Vázquez, E., Ferrer-Miralles, N., 2020. Nanostructured antimicrobial peptides: the last push towards clinics. Biotechnol. Adv. 44, 107603. https://doi.org/10.1016/j.biotechadv.2020.107603.

Castelletto, M.L., Gang, S.S., Hallem, E.A., 2020. Recent advances in functional genomics for parasitic nematodes of mammals. J. Exp. Biol. 223 (Pt Suppl 1), jeb206482. https://doi.org/10.1242/jeb.206482.

Charlier, J., Rinaldi, L., Musella, V., Ploeger, H.W., Chartier, C., Vineer, H.R., Hinney, B., von Samson-Himmelstjerna, G., Băcescu, B., Mickiewicz, M., Mateus, T.L., Martinez-Valladares, M., Quealy, S., Azaizeh, H., Sekovska, B., Akkari, H., Petkevicius, S., Hektoen, L., Höglund, J., Morgan, E., Bartley, D., Claerebout, E., 2020. Initial assessment of the economic burden of major parasitic helminth infections to the ruminant livestock industry in Europe. Prev. Vet. Med. 182, 105103. https://doi.org/10.1016/j.prevetmed.2020.105103.

Charlier, J., Bartley, D.J., Sotiraki, S., Martinez-Valladares, M., Claerebout, E., von Samson-Himmelstjerna, G., Thamsborg, S.M., Hoste, H., Morgan, E.R., Rinaldi, L., 2022. Anthelmintic resistance in ruminants: challenges and solutions. Adv. Parasitol. 115, 171–227. https://doi.org/10.1016/bs.apar.2021.12.002.

Chehayeb, J.F., Robertson, A.P., Martin, R.J., Geary, T.G., 2014. Proteomic analysis of adult *Ascaris suum* fluid compartments and secretory products. PLoS Negl. Trop. Dis. 8 (6), e2939. https://doi.org/10.1371/journal.pntd.0002939.

Chileveru, H.R., Lim, S.A., Chairatana, P., Wommack, A.J., Chiang, I.-L., Nolan, E.M., 2015. Visualizing attack of *Escherichia coli* by the antimicrobial peptide human Defensin 5. Biochemistry 54 (9), 1767–1777. https://doi.org/10.1021/bi501483q.

Chiou, P.P., Chen, M.J., Lin, C.-M., Khoo, J., Larson, J., Holt, R., Leong, J.-A., Thorgarrd, G., Chen, T.T., 2014. Production of homozygous transgenic rainbow trout with enhanced disease resistance. Mar. Biotechnol. 16 (3), 299–308. https://doi.org/10.1007/s10126-013-9550-z.

Choi, H., Yang, Z., Weisshaar, J.C., 2015. Single-cell, real-time detection of oxidative stress induced in *Escherichia coli* by the antimicrobial peptide CM15. Proc. Natl. Acad. Sci. 112 (3), E303–E310. https://doi.org/10.1073/pnas.1417703112.

Choi, H., Yang, Z., Weisshaar, J.C., 2017. Oxidative stress induced in *E. coli* by the human antimicrobial peptide LL-37. PLoS Pathog. 13 (6), e1006481. https://doi.org/10.1371/journal.ppat.1006481.

Cirioni, O., Giacometti, A., Barchiesi, F., Scalise, G., 1998. In-vitro activity of lytic peptides alone and in combination with macrolides and inhibitors of dihydrofolate reductase against *Pneumocystis carinii*. J. Antimicrob. Chemother. 42 (4), 445–451. https://doi.org/10.1093/jac/42.4.445.

Co, J.Y., Margalef-Català, M., Monack, D.M., Amieva, M.R., 2021. Controlling the polarity of human gastrointestinal organoids to investigate epithelial biology and infectious diseases. Nat. Protoc. 16 (11), 5171–5192. https://doi.org/10.1038/s41596-021-00607-0.

Cockx, B., Van Bael, S., Boelen, R., Vandewyer, E., Yang, H., Le, T.A., Dalzell, J.J., Beets, I., Ludwig, C., Lee, J., Temmerman, L., 2023. Mass spectrometry-driven discovery of neuropeptides mediating nictation behavior of nematodes. Mol. Cell. Proteomics 22 (2), 100479. https://doi.org/10.1016/j.mcpro.2022.100479.

Cooper, R., Tsimring, L., Hasty, J., 2018. Microfluidics-based analysis of contact-dependent bacterial interactions. Bio-Protocol 8 (16), e2970. https://doi.org/10.21769/BioProtoc.2970.

Cortés, A., Peachey, L.E., Jenkins, T.P., Scotti, R., Cantacessi, C., 2019. Helminths and microbes within the vertebrate gut—not all studies are created equal. Parasitology 146 (11), 1371–1378. https://doi.org/10.1017/S003118201900088X.

Cortés, A., Wills, J., Su, X., Hewitt, R.E., Robertson, J., Scotti, R., Price, D.R.G., Bartley, Y., McNeilly, T.N., Krause, L., Powell, J.J., Nisbet, A.J., Cantacessi, C., 2020. Infection with the sheep gastrointestinal nematode *Teladorsagia circumcincta* increases luminal pathobionts. Microbiome 8 (1), 60. https://doi.org/10.1186/s40168-020-00818-9.

Couillault, C., Pujol, N., Reboul, J., Sabatier, L., Guichou, J.-F., Kohara, Y., Ewbank, J.J., 2004. TLR-independent control of innate immunity in *Caenorhabditis elegans* by the TIR domain adaptor protein TIR-1, an ortholog of human SARM. Nat. Immunol. 5 (5), 488–494. https://doi.org/10.1038/ni1060.

da Silva, B.R., de Freitas, V.A.A., Carneiro, V.A., Arruda, F.V.S., Lorenzón, E.N., de Aguiar, A.S.W., Cilli, E.M., Cavada, B.S., Teixeira, E.H., 2013. Antimicrobial activity of the synthetic peptide Lys-a1 against oral streptococci. Peptides 42, 78–83. https://doi.org/10.1016/j.peptides.2012.12.001.

Deng, T., Ge, H., He, H., Liu, Y., Zhai, C., Feng, L., Yi, L., 2017. The heterologous expression strategies of antimicrobial peptides in microbial systems. Protein Expr. Purif. 140, 52–59. https://doi.org/10.1016/j.pep.2017.08.003.

Dieterich, C., Clifton, S.W., Schuster, L.N., Chinwalla, A., Delehaunty, K., Dinkelacker, I., Fulton, L., Fulton, R., Godfrey, J., Minx, P., Mitreva, M., Roeseler, W., Tian, H., Witte, H., Yang, S.-P., Wilson, R.K., Sommer, R.J., 2008. The *Pristionchus pacificus* genome provides a unique perspective on nematode lifestyle and parasitism. Nat. Genet. 40 (10), 1193–1198. https://doi.org/10.1038/ng.227.

Doyle, S.R., 2022. Improving helminth genome resources in the post-genomic era. Trends Parasitol. 38 (10), 831–840. https://doi.org/10.1016/j.pt.2022.06.002.

Drurey, C., Lindholm, H.T., Coakley, G., Poveda, M.C., Löser, S., Doolan, R., Gerbe, F., Jay, P., Harris, N., Oudhoff, M.J., Maizels, R.M., 2021. Intestinal epithelial tuft cell induction is negated by a murine helminth and its secreted products. J. Exp. Med. 219 (1), e20211140. https://doi.org/10.1084/jem.20211140.

Ducarmon, Q.R., Hoogerwerf, M.A., Janse, J.J., Geelen, A.R., Koopman, J.P.R., Zwittink, R.D., Goeman, J.J., Kuijper, E.J., Roestenberg, M., 2020. Dynamics of the bacterial gut microbiota during controlled human infection with *Necator americanus* larvae. Gut Microbes 12 (1), 1–15. https://doi.org/10.1080/19490976.2020.1840764.

Duque-Correa, M.A., Schreiber, F., Rodgers, F.H., Goulding, D., Forrest, S., White, R., Buck, A., Grencis, R.K., Berriman, M., 2020. Development of caecaloids to study host–pathogen interactions: new insights into immunoregulatory functions of *Trichuris muris* extracellular vesicles in the caecum. Int. J. Parasitol. 50 (9), 707–718. https://doi.org/10.1016/j.ijpara.2020.06.001.

Duque-Correa, M.A., Goulding, D., Rodgers, F.H., Gillis, J.A., Cormie, C., Rawlinson, K. A., Bancroft, A.J., Bennett, H.M., Lotkowska, M.E., Reid, A.J., Speak, A.O., Scott, P., Redshaw, N., Tolley, C., McCarthy, C., Brandt, C., Sharpe, C., Ridley, C., Moya, J., Carneiro, C.M., Starborg, T., Hayes, K.S., Holroyd, N., Sanders, M., Thornton, D.J., Grencis, R.K., Berriman, M., 2022. Defining the early stages of intestinal colonisation by whipworms. Nat. Commun. 13 (1), 1725. https://doi.org/10.1038/s41467-022-29334-0.

El-Baky, N.A., Elkhawaga, M.A., Abdelkhalek, E.S., Sharaf, M.M., Redwan, E.M., Kholef, H.R., 2020. De novo expression and antibacterial potential of four lactoferricin peptides in cell-free protein synthesis system. Biotechnol. Rep. 29, e00583. https://doi.org/10.1016/j.btre.2020.e00583.

Else, K.J., Keiser, J., Holland, C.V., Grencis, R.K., Sattelle, D.B., Fujiwara, R.T., Bueno, L. L., Asaolu, S.O., Sowemimo, O.A., Cooper, P.J., 2020. Whipworm and roundworm infections. Nat. Rev. Dis. Primers 6 (1), 1–23. https://doi.org/10.1038/s41572-020-0171-3.

Ewbank, J.J., Zugasti, O., 2011. *C. elegans*: model host and tool for antimicrobial drug discovery. Dis. Model. Mech. 4 (3), 300–304. https://doi.org/10.1242/dmm.006684.

Faber, M.N., Smith, D., Price, D.R.G., Steele, P., Hildersley, K.A., Morrison, L.J., Mabbott, N.A., Nisbet, A.J., McNeilly, T.N., 2022. Development of bovine gastric organoids as a novel in vitro model to study host-parasite interactions in gastrointestinal nematode infections. Front. Cell. Infect. Microbiol. 12, 904606. https://doi.org/10.3389/fcimb.2022.904606.

Fantner, G.E., Barbero, R.J., Gray, D.S., Belcher, A.M., 2010. Kinetics of antimicrobial peptide activity measured on individual bacterial cells using high-speed atomic force microscopy. Nat. Nanotechnol. 5 (4), 280–285. https://doi.org/10.1038/nnano.2010.29.

Faust, J.E., Yang, P.-Y., Huang, H.W., 2017. Action of antimicrobial peptides on bacterial and lipid membranes: a direct comparison. Biophys. J. 112 (8), 1663–1672. https://doi.org/10.1016/j.bpj.2017.03.003.

Ferguson, A.A., Rossi, H.L., Herbert, D.R., 2024. The secretome of adult murine hookworms is shaped by host expression of STAT6. Parasite Immunol. 46 (7), e13056. https://doi.org/10.1111/pim.13056.

Fofanova, T.Y., Karandikar, U.C., Auchtung, J.M., Wilson, R.L., Valentin, A.J., Britton, R. A., Grande-Allen, K.J., Estes, M.K., Hoffman, K., Ramani, S., Stewart, C.J., Petrosino, J.F., 2024. A novel system to culture human intestinal organoids under physiological oxygen content to study microbial-host interaction. PLoS One 19 (7), e0300666. https://doi.org/10.1371/journal.pone.0300666.

Gazit, E., Boman, A., Boman, H.G., Shai, Y., 1995. Interaction of the mammalian antibacterial peptide Cecropin P1 with phospholipid vesicles. Biochemistry 34 (36), 11479–11488. https://doi.org/10.1021/bi00036a021.

Gazit, E., Miller, I.R., Biggin, P.C., Sansom, M.S.P., Shai, Y., 1996. Structure and orientation of the mammalian antibacterial peptide Cecropin P1 within phospholipid membranes. J. Mol. Biol. 258 (5), 860–870. https://doi.org/10.1006/jmbi.1996.0293.

Gerdol, M., De Moro, G., Manfrin, C., Venier, P., Pallavicini, A., 2012. Big defensins and mytimacins, new AMP families of the Mediterranean mussel *Mytilus galloprovincialis*. Dev. Comp. Immunol. 36 (2), 390–399. https://doi.org/10.1016/j.dci.2011.08.003.

Giacometti, A., Cirioni, O., Barchiesi, F., Del Prete, M.S., Fortuna, M., Caselli, F., Scalise, G., 2000. In vitro susceptibility tests for cationic peptides: comparison of broth microdilution methods for bacteria that grow aerobically. Antimicrob. Agents Chemother. 44 (6), 1694–1696. https://doi.org/10.1128/aac.44.6.1694-1696.2000.

Gilleard, J.S., Kotze, A.C., Leathwick, D., Nisbet, A.J., McNeilly, T.N., Besier, B., 2021. A journey through 50 years of research relevant to the control of gastrointestinal nematodes in ruminant livestock and thoughts on future directions. Int. J. Parasitol. 51 (13), 1133–1151. https://doi.org/10.1016/j.ijpara.2021.10.007.

Gillet, L.C., Navarro, P., Tate, S., Röst, H., Selevsek, N., Reiter, L., Bonner, R., Aebersold, R., 2012. Targeted data extraction of the MS/MS spectra generated by data-independent acquisition: A new concept for consistent and accurate proteome analysis. Mol. Cell. Proteomics 11 (6). https://doi.org/10.1074/mcp.O111.016717.

Gomes, A., Byregowda, S., B M, V, Vinayagamurthy, B., 2016. An overview of heterologous expression host systems for the production of recombinant proteins. Adv. Anim. Vet. Sci. 4, 346–356. https://doi.org/10.14737/journal.aavs/2016/4.7.346.356.

Goud, G.N., Bottazzi, M.E., Zhan, B., Mendez, S., Deumic, V., Plieskatt, J., Liu, S., Wang, Y., Bueno, L., Fujiwara, R., Samuel, A., Ahn, S.Y., Solanki, M., Asojo, O.A., Wang, J., Bethony, J.M., Loukas, A., Roy, M., Hotez, P.J., 2005. Expression of the *Necator americanus* hookworm larval antigen *Na*-ASP-2 in *Pichia pastoris* and purification of the recombinant protein for use in human clinical trials. Vaccine 23 (39), 4754–4764. https://doi.org/10.1016/j.vaccine.2005.04.040.

Gu, J., Gao, B., Zhu, S., 2018. Characterization of bi-domain drosomycin-type antifungal peptides in nematodes: an example of convergent evolution. Dev. Comp. Immunol. 87, 90–97. https://doi.org/10.1016/j.dci.2018.06.005.

Guo, C., Huang, Y., Cong, P., Liu, X., Chen, Y., He, Z., 2014. Cecropin P1 inhibits porcine reproductive and respiratory syndrome virus by blocking attachment. BMC Microbiol. 14 (1), 273. https://doi.org/10.1186/s12866-014-0273-8.

Guryanova, S.V., Balandin, S.V., Belogurova-Ovchinnikova, O., Yu, Ovchinnikova, T.V., 2023. Marine invertebrate antimicrobial peptides and their potential as novel peptide antibiotics. Mar. Drugs 21 (10), 503. https://doi.org/10.3390/md21100503.

Herbst, R., Marciano-Cabral, F., Leippe, M., 2004. Antimicrobial and pore-forming peptides of free-living and potentially highly pathogenic *Naegleria fowleri* are released from the same precursor molecule. J. Biol. Chem. 279 (25), 25955–25958. https://doi.org/10.1074/jbc.M401965200.

Hoeckendorf, A., Leippe, M., 2012. SPP-3, a saposin-like protein of *Caenorhabditis elegans*, displays antimicrobial and pore-forming activity and is located in the intestine and in one head neuron. Dev. Comp. Immunol. 38 (1), 181–186. https://doi.org/10.1016/j.dci.2012.05.007.

Hoelscher, M.P., Forner, J., Calderone, S., Krämer, C., Taylor, Z., Loiacono, F.V., Agrawal, S., Karcher, D., Moratti, F., Kroop, X., Bock, R., 2022. Expression strategies for the efficient synthesis of antimicrobial peptides in plastids. Nat. Commun. 13 (1), 5856. https://doi.org/10.1038/s41467-022-33516-1.

Huan, Y., Kong, Q., Mou, H., Yi, H., 2020. Antimicrobial peptides: classification, design, application and research progress in multiple fields. Front. Microbiol. 11, 582779.. https://doi.org/10.3389/fmicb.2020.582779.

Hussain, W., 2022. sAMP-PFPDeep: improving accuracy of short antimicrobial peptides prediction using three different sequence encodings and deep neural networks. Brief. Bioinform. 23 (1), bbab487. https://doi.org/10.1093/bib/bbab487.

Implen, 2020, November 10. The OD600 Basics | Best OD600 Tool To Generate Microbial Growth Curves. https://www.implen.de/od600-diluphotometer/od600/.

Irvine, A., McKenzie, D., McCoy, C.J., Graham, R.L.J., Graham, C., Huws, S.A., Atkinson, L.E., Mousley, A., 2023a. Novel integrated computational AMP discovery approaches highlight diversity in the helminth AMP repertoire. PLoS Pathog. 19 (7), e1011508. https://doi.org/10.1371/journal.ppat.1011508.

Irvine, A., Huws, S.A., Atkinson, L.E., Mousley, A., 2023b. Exploring the antimicrobial peptidome of nematodes through phylum-spanning in silico analyses highlights novel opportunities for pathogen control. PLoS Negl. Trop. Dis. 17 (9), e0011618. https://doi.org/10.1371/journal.pntd.0011618.

Jarecki, J.L., Frey, B.L., Smith, L.M., Stretton, A.O., 2011. Discovery of neuropeptides in the nematode *Ascaris suum* by database mining and tandem mass spectrometry. J. Proteome Res. 10 (7), 3098–3106. https://doi.org/10.1021/pr2001176.

Jenkins, T., Formenti, F., Castro, C., Piubelli, C., Perandin, F., Buonfrate, D., Otranto, D., Griffin, J., Krause, L., Bisoffi, Z., Cantacessi, C., 2018. A comprehensive analysis of the faecal microbiome and metabolome of *Strongyloides stercoralis* infected volunteers from a non-endemic area. Sci. Rep. 8, 15651. https://doi.org/10.1038/s41598-018-33937-3.

Jia, T.-W., Melville, S., Utzinger, J., King, C.H., Zhou, X.-N., 2012. Soil-transmitted helminth reinfection after drug treatment: A systematic review and meta-analysis. PLoS Negl. Trop. Dis. 6 (5), e1621. https://doi.org/10.1371/journal.pntd.0001621.

Kandaswamy, K., Liew, T.H., Wang, C.Y., Huston-Warren, E., Meyer-Hoffert, U., Hultenby, K., Schröder, J.M., Caparon, M.G., Normark, S., Henriques-Normark, B., Hultgren, S.J., Kline, K.A., 2013. Focal targeting by human β-defensin 2 disrupts localized virulence factor assembly sites in *Enterococcus faecalis*. Proc. Natl. Acad. Sci. 110 (50), 20230–20235. https://doi.org/10.1073/pnas.1319066110.

Karo-Atar, D., Ouladan, S., Javkar, T., Joumier, L., Matheson, M.K., Merritt, S., Westfall, S., Rochette, A., Gentile, M.E., Fontes, G., Fonseca, G.J., Parisien, M., Diatchenko, L., von Moltke, J., Malleshaiah, M., Gregorieff, A., King, I.L., 2022. Helminth-induced reprogramming of the stem cell compartment inhibits type 2 immunity. J. Exp. Med. 219 (9), e20212311. https://doi.org/10.1084/jem.20212311.

Karve, S.S., Pradhan, S., Ward, D.V., Weiss, A.A., 2017. Intestinal organoids model human responses to infection by commensal and Shiga toxin producing *Escherichia coli*. PLoS One 12 (6), e0178966. https://doi.org/10.1371/journal.pone.0178966.

Kato, Y., 1995. Humoral defense of the nematode *Ascaris suum*: antibacterial, bacteriolytic and agglutinating activities in the body fluid. Zool. Sci. 12 (2), 225–230. https://doi.org/10.2108/zsj.12.225.

Kato, Y., Komatsu, S., 1996. ASABF, a novel cysteine-rich antibacterial peptide isolated from the nematode *Ascaris suum*: purification, primary structure, and molecular cloning of cDNA. J. Biol. Chem. 271 (48), 30493–30498. https://doi.org/10.1074/jbc.271.48.30493.

Kato, Y., Aizawa, T., Hoshino, H., Kawano, K., Nitta, K., Zhang, H., 2002. *Abf-1* and *Abf-2*, ASABF-type antimicrobial peptide genes in *Caenorhabditis elegans*. Biochem. J. 361 (2), 221–230. https://doi.org/10.1042/bj3610221.

Kim, J.Y., Kim, Y.-G., Lee, G.M., 2012. CHO cells in biotechnology for production of recombinant proteins: Current state and further potential. Appl. Microbiol. Biotechnol. 93 (3), 917–930. https://doi.org/10.1007/s00253-011-3758-5.

Klubthawee, N., Adisakwattana, P., Hanpithakpong, W., Somsri, S., Aunpad, R., 2020. A novel, rationally designed, hybrid antimicrobial peptide, inspired by cathelicidin and aurein, exhibits membrane-active mechanisms against *Pseudomonas aeruginosa*. Sci. Rep. 10, 9117. https://doi.org/10.1038/s41598-020-65688-5.

Kupritz, J., Angelova, A., Nutman, T.B., Gazzinelli-Guimaraes, P.H., 2021. Helminth-induced human gastrointestinal dysbiosis: A systematic review and meta-analysis reveals insights into altered taxon diversity and microbial gradient collapse. mBio 12 (6), e0289281. https://doi.org/10.1128/mBio.02890-21.

Kyriazakis, I., Arndt, C., Aubry, A., Charlier, J., Ezenwa, V.O., Godber, O.F., Krogh, M., Mostert, P.F., Orsel, K., Robinson, M.W., Ryan, F.S., Skuce, P.J., Takahashi, T., van Middelaar, C.E., Vigors, S., Morgan, E.R., 2024. Improve animal health to reduce livestock emissions: quantifying an open goal. Proc. R. Soc. B Biol. Sci. 291 (2027), 20240675. https://doi.org/10.1098/rspb.2024.0675.

Lacroix-Lamandé, S., Bernardi, O., Pezier, T., Barilleau, E., Burlaud-Gaillard, J., Gagneux, A., Velge, P., Wiedemann, A., 2023. Differential *Salmonella Typhimurium* intracellular replication and host cell responses in caecal and ileal organoids derived from chicken. Vet. Res. 54 (1), 63. https://doi.org/10.1186/s13567-023-01189-3.

Lee, J.Y., Boman, A., Sun, C.X., Andersson, M., Jörnvall, H., Mutt, V., Boman, H.G., 1989. Antibacterial peptides from pig intestine: isolation of a mammalian cecropin. Proc. Natl. Acad. Sci. 86 (23), 9159. https://doi.org/10.1073/pnas.86.23.9159.

Lezi, E., Zhou, T., Koh, S., Chuang, M., Sharma, R., Pujol, N., Chisholm, A.D., Eroglu, C., Matsunami, H., Yan, D., 2018. An antimicrobial peptide and its neuronal receptor regulate dendrite degeneration in aging and infection. Neuron 97 (1), 125–138.e5. https://doi.org/10.1016/j.neuron.2017.12.001.

Li, Y., 2011. Recombinant production of antimicrobial peptides in *Escherichia coli*: A review. Protein Expr. Purif. 80 (2), 260–267. https://doi.org/10.1016/j.pep.2011.08.001.

Li, R.W., Li, W., Sun, J., Yu, P., Baldwin, R.L., Urban, J.F., 2016. The effect of helminth infection on the microbial composition and structure of the caprine abomasal microbiome. Sci. Rep. 6, 20606. https://doi.org/10.1038/srep20606.

Li, J.-F., Zhang, J.-X., Li, G., Xu, Y.-Y., Lu, K., Wang, Z.-G., Liu, J.-P., 2020. Antimicrobial activity and mechanism of peptide CM4 against *Pseudomonas aeruginosa*. Food Funct. 11 (8), 7245–7254. https://doi.org/10.1039/D0FO01031F.

Lim, M.P., Firdaus-Raih, M., Nathan, S., 2016. Nematode peptides with host-directed anti-inflammatory activity rescue *Caenorhabditis elegans* from a *Burkholderia pseudomallei* infection. Front. Microbiol. 7, 1436. https://doi.org/10.3389/fmicb.2016.01436.

Lin, T.-T., Yang, L.-Y., Lu, I.-H., Cheng, W.-C., Hsu, Z.-R., Chen, S.-H., Lin, C.-Y., 2021. AI4AMP: an antimicrobial peptide predictor using physicochemical property-based encoding method and deep learning. mSystems 6 (6), e0029921. https://doi.org/10.1128/mSystems.00299-21.

Logan, J., Pearson, M.S., Manda, S.S., Choi, Y.-J., Field, M., Eichenberger, R.M., Mulvenna, J., Nagaraj, S.H., Fujiwara, R.T., Gazzinelli-Guimaraes, P., Bueno, L., Mati, V., Bethony, J.M., Mitreva, M., Sotillo, J., Loukas, A., 2020. Comprehensive analysis of the secreted proteome of adult *Necator americanus* hookworms. PLoS Negl. Trop. Dis. 14 (5), e0008237. https://doi.org/10.1371/journal.pntd.0008237.

Loukas, A., Hotez, P.J., Diemert, D., Yazdanbakhsh, M., McCarthy, J.S., Correa-Oliveira, R., Croese, J., Bethony, J.M., 2016. Hookworm infection. Nat. Rev. Dis. Prim. 2, 16088. https://doi.org/10.1038/nrdp.2016.88.

Loukas, A., Maizels, R.M., Hotez, P.J., 2021. The yin and yang of human soil-transmitted helminth infections. Int. J. Parasitol. 51 (13), 1243–1253. https://doi.org/10.1016/j.ijpara.2021.11.001.

Ludwig, C., Gillet, L., Rosenberger, G., Amon, S., Collins, B.C., Aebersold, R., 2018. Data-independent acquisition-based SWATH-MS for quantitative proteomics: a tutorial. Mol. Syst. Biol. 14 (8), e8126. https://doi.org/10.15252/msb.20178126.

Maizels, R.M., 2020. Regulation of immunity and allergy by helminth parasites. Allergy 75 (3), 524–534. https://doi.org/10.1111/all.13944.

Matsuzaki, K., 2019. Membrane Permeabilization Mechanisms. Adv. Exp. Med. Biol. 1117, 9–16. https://doi.org/10.1007/978-981-13-3588-4_2.

McCoy, C.J., Warnock, N.D., Atkinson, L.E., Atcheson, E., Martin, R.J., Robertson, A.P., Maule, A.G., Marks, N.J., Mousley, A., 2015. RNA interference in adult *Ascaris suum* – an opportunity for the development of a functional genomics platform that supports organism-, tissue- and cell-based biology in a nematode parasite. Int. J. Parasitol. 45 (11), 673–678. https://doi.org/10.1016/j.ijpara.2015.05.003.

McDonald, E.F., Jones, T., Plate, L., Meiler, J., Gulsevin, A., 2022. Benchmarking AlphaFold2 on peptide structure prediction. Structure 31 (1), 111–119. https://doi.org/10.1016/j.str.2022.11.012.

McVeigh, P., Alexander-Bowman, S., Veal, E., Mousley, A., Marks, N.J., Maule, A.G., 2008. Neuropeptide-like protein diversity in phylum *Nematoda*. Int. J. Parasitol. 38 (13), 1493–1503. https://doi.org/10.1016/j.ijpara.2008.05.006.

Meher, P., Sahu, T., Saini, V., Rao, A., 2017. Predicting antimicrobial peptides with improved accuracy by incorporating the compositional, physico-chemical and structural features into Chou's general PseAAC. Sci. Rep. 7, 42362. https://doi.org/10.1038/srep42362.

Meier, F., Brunner, A.-D., Koch, S., Koch, H., Lubeck, M., Krause, M., Goedecke, N., Decker, J., Kosinski, T., Park, M.A., Bache, N., Hoerning, O., Cox, J., Räther, O., Mann, M., 2018. Online parallel accumulation-serial fragmentation (PASEF) with a novel trapped ion mobility mass spectrometer. Mol. Cell. Proteomics 17 (12), 2534–2545. https://doi.org/10.1074/mcp.TIR118.000900.

Meier, F., Brunner, A.-D., Frank, M., Ha, A., Bludau, I., Voytik, E., Kaspar-Schoenefeld, S., Lubeck, M., Raether, O., Bache, N., Aebersold, R., Collins, B.C., Röst, H.L., Mann, M., 2020. diaPASEF: parallel accumulation-serial fragmentation combined with data-independent acquisition. Nat. Methods 17 (12), 1229–1236. https://doi.org/10.1038/s41592-020-00998-0.

Michalek, M., Leippe, M., 2015. Mechanistic insights into the lipid interaction of an ancient saposin-like protein. Biochemistry 54 (9), 1778–1786. https://doi.org/10.1021/acs.biochem.5b00094.

Midha, A., Schlosser, J., Hartmann, S., 2017. Reciprocal interactions between nematodes and their microbial environments. Front. Cell. Infect. Microbiol. 7, 144. https://doi.org/10.3389/fcimb.2017.00144.

Midha, A., Janek, K., Niewienda, A., Henklein, P., Guenther, S., Serra, D.O., Schlosser, J., Hengge, R., Hartmann, S., 2018. The intestinal roundworm *Ascaris suum* releases antimicrobial factors which interfere with bacterial growth and biofilm formation. Front. Cell. Infect. Microbiol. 8. https://doi.org/10.3389/fcimb.2018.00271.

Minaba, M., Ueno, S., Pillai, A., Kato, Y., 2009. Evolution of ASABF (*Ascaris suum* antibacterial factor)-type antimicrobial peptides in nematodes: putative rearrangement of disulfide bonding patterns. Dev. Comp. Immunol. 33 (11), 1147–1150. https://doi.org/10.1016/j.dci.2009.06.011.

Mladineo, I., Rončević, T., Gerdol, M., Tossi, A., 2023. Helminthic host defense peptides: using the parasite to defend the host. Trends Parasitol. 39 (5), 345–357. https://doi.org/10.1016/j.pt.2023.02.004.

Moghal, M.M.R., Hossain, F., Yamazaki, M., 2020. Action of antimicrobial peptides and cell-penetrating peptides on membrane potential revealed by the single GUV method. Biophys. Rev. 12 (2), 339. https://doi.org/10.1007/s12551-020-00662-z.

Moreno, Y., Gros, P.-P., Tam, M., Segura, M., Valanparambil, R., Geary, T.G., Stevenson, M.M., 2011. Proteomic analysis of excretory-secretory products of *Heligmosomoides polygyrus* assessed with next-generation sequencing transcriptomic information. PLoS Negl. Trop. Dis. 5 (10), e1370. https://doi.org/10.1371/journal.pntd.0001370.

Müller, H., Salzig, D., Czermak, P., 2015. Considerations for the process development of insect-derived antimicrobial peptide production. Biotechnol. Prog. 31 (1), 1–11. https://doi.org/10.1002/btpr.2002.

Münzker, L., Oddo, A., Hansen, P.R., 2017. Chemical Synthesis of Antimicrobial Peptides. In: Hansen, P. (Ed.), Antimicrobial Peptides, Methods Mol. Biol., 1548. Humana Press, New York, NY, pp. 35–49. https://doi.org/10.1007/978-1-4939-6737-7_3.

Mylonakis, E., Podsiadlowski, L., Muhammed, M., Vilcinskas, A., 2016. Diversity, evolution and medical applications of insect antimicrobial peptides. Philos. Trans. R. Soc. Lond. Ser. B Biol. Sci. 371 (1695), 20150290. https://doi.org/10.1098/rstb.2015.0290.

Nash, T.J., Morris, K.M., Mabbott, N.A., Vervelde, L., 2021. Inside-out chicken enteroids with leukocyte component as a model to study host-pathogen interactions. Commun. Biol. 4 (1), 377. https://doi.org/10.1038/s42003-021-01901-z.

Nisbet, A.J., Meeusen, E.N., González, J.F., Piedrafita, D.M., 2016. Immunity to *Haemonchus contortus* and vaccine development. Adv. Parasitol. 93, 353–396. https://doi.org/10.1016/bs.apar.2016.02.011.

Nisbet, A.J., McNeilly, T.N., Price, D.R.G., Oliver, E.M., Bartley, Y., Mitchell, M., Palarea-Albaladejo, J., Matthews, J.B., 2019. The rational simplification of a recombinant cocktail vaccine to control the parasitic nematode *Teladorsagia circumcincta*. Int. J. Parasitol. 49 (3–4), 257–265. https://doi.org/10.1016/j.ijpara.2018.10.006.

Nixon, S.A., Welz, C., Woods, D.J., Costa-Junior, L., Zamanian, M., Martin, R.J., 2020. Where are all the anthelmintics? Challenges and opportunities on the path to new anthelmintics. Int. J. Parasitol. Drugs Drug Resist. 14, 8–16. https://doi.org/10.1016/j.ijpddr.2020.07.001.

Oliveira, K.B.S., de Leite, M.L., Rodrigues, G.R., Duque, H.M., da Costa, R.A., Cunha, V. A., de Costa, L.S.L., da Cunha, N.B., Franco, O.L., Dias, S.C., 2020. Strategies for recombinant production of antimicrobial peptides with pharmacological potential. Expert. Rev. Clin. Pharmacol. 13 (4), 367–390. https://doi.org/10.1080/17512433.2020.1764347.

Peachey, L.E., Jenkins, T.P., Cantacessi, C., 2017. This gut ain't big enough for both of us. Or is it? Helminth–microbiota interactions in veterinary species. Trends Parasitol. 33 (8), 619–632. https://doi.org/10.1016/j.pt.2017.04.004.

Pillai, A., Ueno, S., Zhang, H., Kato, Y., 2003. Induction of ASABF (*Ascaris suum* antibacterial factor)-type antimicrobial peptides by bacterial injection: novel members of ASABF in the nematode *Ascaris suum*. Biochem. J. 371 (Pt 3), 663–668. https://doi.org/10.1042/BJ20021948.

Pillai, A., Ueno, S., Zhang, H., Lee, J.M., Kato, Y., 2005. Cecropin P1 and novel nematode cecropins: A bacteria-inducible antimicrobial peptide family in the nematode *Ascaris suum*. Biochem. J. 390 (Pt 1), 207. https://doi.org/10.1042/BJ20050218.

Pina, A., Batalha, Í., Roque, A., 2014. Affinity tags in protein purification and peptide enrichment: an overview. In: Methods in Molecular Biology (Clifton, N.J.), 1129, pp. 147–168. https://doi.org/10.1007/978-1-62703-977-2_14.

Pirtskhalava, M., Amstrong, A.A., Grigolava, M., Chubinidze, M., Alimbarashvili, E., Vishnepolsky, B., Gabrielian, A., Rosenthal, A., Hurt, D.E., Tartakovsky, M., 2021. DBAASP v3: database of antimicrobial/cytotoxic activity and structure of peptides as a resource for development of new therapeutics. Nucleic Acids Res. 49 (D1), D288–D297. https://doi.org/10.1093/nar/gkaa991.

Pleguezuelos-Manzano, C., Puschhof, J., van den Brink, S., Geurts, V., Beumer, J., Clevers, H., 2020. Establishment and culture of human intestinal organoids derived from adult stem cells. Curr. Protoc. Immunol. 130 (1), e106. https://doi.org/10.1002/cpim.106.

Porto, W.F., Pires, A.S., Franco, O.L., 2017. Computational tools for exploring sequence databases as a resource for antimicrobial peptides. Biotechnol. Adv. 35 (3), 337–349. https://doi.org/10.1016/j.biotechadv.2017.02.001.

Puchner, K.P., Bottazzi, M.E., Periago, V., Grobusch, M., Maizels, R., McCarthy, J., Lee, B., Gaspari, E., Diemert, D., Hotez, P., 2024. Vaccine value profile for hookworm. Vaccine 42 (19, Supplement 1), S25–S41. https://doi.org/10.1016/j.vaccine.2023.05.013.

Pujol, N., Davis, P.A., Ewbank, J.J., 2012. The origin and function of anti-fungal peptides in *C. elegans*: open questions. Front. Immunol. 3, 237. https://doi.org/10.3389/fimmu.2012.00237.

Ramazi, S., Mohammadi, N., Allahverdi, A., Khalili, E., Abdolmaleki, P., 2022. A review on antimicrobial peptides databases and the computational tools. Database J. Biol. Databases Curation 2022, baac011. https://doi.org/10.1093/database/baac011.

Rangarajan, N., Bakshi, S., Weisshaar, J.C., 2013. Localized permeabilization of *E. coli* membranes by the antimicrobial peptide Cecropin A. Biochemistry 52 (38), 6584–6594. https://doi.org/10.1021/bi400785j.

Rausch, S., Midha, A., Kuhring, M., Affinass, N., Radonic, A., Kühl, A.A., Bleich, A., Renard, B.Y., Hartmann, S., 2018. Parasitic nematodes exert antimicrobial activity and benefit from microbiota-driven support for host immune regulation. Front. Immunol. 9, 2282. https://doi.org/10.3389/fimmu.2018.02282.

Remans, K., Lebendiker, M., Abreu, C., Maffei, M., Sellathurai, S., May, M.M., Vaněk, O., de Marco, A., 2022. Protein purification strategies must consider downstream applications and individual biological characteristics. Microb. Cell Factories 21, 52. https://doi.org/10.1186/s12934-022-01778-5.

Rödelsperger, C., Sommer, R.J., 2011. Computational archaeology of the *Pristionchus pacificus* genome reveals evidence of horizontal gene transfers from insects. BMC Evol. Biol. 11 (1), 239. https://doi.org/10.1186/1471-2148-11-239.

Roeder, T., Stanisak, M., Gelhaus, C., Bruchhaus, I., Grötzinger, J., Leippe, M., 2010. Caenopores are antimicrobial peptides in the nematode *Caenorhabditis elegans* instrumental in nutrition and immunity. Dev. Comp. Immunol. 34 (2), 203–209. https://doi.org/10.1016/j.dci.2009.09.010.

Rončević, T., Gerdol, M., Mardirossian, M., Maleš, M., Cvjetan, S., Benincasa, M., Maravić, A., Gajski, G., Krce, L., Aviani, I., Hrabar, J., Trumbić, Ž., Derks, M., Pallavicini, A., Weingarth, M., Zoranić, L., Tossi, A., Mladineo, I., 2022. Anisaxins, helical antimicrobial peptides from marine parasites, kill resistant bacteria by lipid extraction and membrane disruption. Acta Biomater. 146, 131–144. https://doi.org/10.1016/j.actbio.2022.04.025.

Rooney, J., Northcote, H.M., Williams, T.I., Cortés, A., Cantacessi, C., Morphew, R., 2022a. Parasitic helminths and the host microbiome—A missing 'extracellular vesicle-sized' link? Trends Parasitol. 38 (9), 737–747. https://doi.org/10.1016/j.pt.2022.06.003.

Rooney, J., Williams, T.L., Northcote, H.M., Frankl, F.E.K., Price, D.R.G., Nisbet, A.J., Morphew, R.M., Cantacessi, C., 2022b. Excretory-secretory products from the brown stomach worm, *Teladorsagia circumcincta*, exert antimicrobial activity in in vitro growth assays. Parasit. Vectors 15 (1), 354. https://doi.org/10.1186/s13071-022-05443-z.

Rooney, J., Rivera-de-Torre, E., Li, R., Mclean, K., Price, D.R.G., Nisbet, A.J., Laustsen, A. H., Jenkins, T.P., Hofmann, A., Bakshi, S., Zarkan, A., Cantacessi, C., 2024. Structural and functional analyses of nematode-derived antimicrobial peptides support the occurrence of direct mechanisms of worm-microbiota interactions. Comput. Struct. Biotechnol. J. 23, 1522–1533. https://doi.org/10.1016/j.csbj.2024.04.019.

Sánchez-Trasviña, C., Flores-Gatica, M., Enriquez-Ochoa, D., Rito-Palomares, M., Mayolo-Deloisa, K., 2021. Purification of modified therapeutic proteins available on the market: an analysis of chromatography-based strategies. Front. Bioeng. Biotechnol. 9, 717326. https://doi.org/10.3389/fbioe.2021.717326.

Scheinpflug, K., Krylova, O., Nikolenko, H., Thurm, C., Dathe, M., 2015. Evidence for a novel mechanism of antimicrobial action of a cyclic R-,W-rich hexapeptide. PLoS One 10 (4), e0125056. https://doi.org/10.1371/journal.pone.0125056.

Sidorczuk, K., Gagat, P., Pietluch, F., Kała, J., Rafacz, D., Bąkała, L., Słowik, J., Kolenda, R., Rödiger, S., Fingerhut, L.C.H.W., Cooke, I.R., Mackiewicz, P., Burdukiewicz, M., 2022. Benchmarks in antimicrobial peptide prediction are biased due to the selection of negative data. Brief. Bioinform. 23 (5), bbac343. https://doi.org/10.1093/bib/bbac343.

Singh, N., Bose, K., 2022. Introduction to recombinant protein purification. In: Bose, K. (Ed.), Textbook on Cloning, Expression and Purification of Recombinant Proteins. Springer Nature, pp. 115–140. https://doi.org/10.1007/978-981-16-4987-5_5.

Sinner, M.P., Masurat, F., Ewbank, J.J., Pujol, N., Bringmann, H., 2021. Innate immunity promotes sleep through epidermal antimicrobial peptides. Curr. Biol. 31 (3), 564–577 e12. https://doi.org/10.1016/j.cub.2020.10.076.

Smith, S.K., Nisbet, A.J., Meikle, L.I., Inglis, N.F., Sales, J., Beynon, R.J., Matthews, J.B., 2009. Proteomic analysis of excretory/secretory products released by *Teladorsagia circumcincta* larvae early post-infection. Parasite Immunol. 31 (1), 10–19. https://doi.org/10.1111/j.1365-3024.2008.01067.x.

Smith, D., Price, D.R.G., Burrells, A., Faber, M.N., Hildersley, K.A., Chintoan-Uta, C., Chapuis, A.F., Stevens, M., Stevenson, K., Burgess, S.T.G., Innes, E.A., Nisbet, A.J., McNeilly, T.N., 2021. The development of ovine gastric and intestinal organoids for studying ruminant host-pathogen interactions. Front. Cell. Infect. Microbiol. 11, 733811. https://doi.org/10.3389/fcimb.2021.733811.

Smith, D., Price, D.R.G., Faber, M.N., Chapuis, A.F., McNeilly, T.N., 2022. Advancing animal health and disease research in the lab with three-dimensional cell culture systems. Vet. Rec. 191 (1), e1528. https://doi.org/10.1002/vetr.1528.

Soblik, H., Younis, A.E., Mitreva, M., Renard, B.Y., Kirchner, M., Geisinger, F., Steen, H., Brattig, N.W., 2011. Life cycle stage-resolved proteomic analysis of the excretome/secretome from *Strongyloides ratti*—identification of stage-specific proteases. Mol. Cell. Proteomics 10 (12). https://doi.org/10.1074/mcp.M111.010157.

Sochacki, K.A., Barns, K.J., Bucki, R., Weisshaar, J.C., 2011. Real-time attack on single *Escherichia coli* cells by the human antimicrobial peptide LL-37. Proc. Natl. Acad. Sci. 108 (16), E77–E81. https://doi.org/10.1073/pnas.1101130108.

Sorobetea, D., Svensson-Frej, M., Grencis, R., 2018. Immunity to gastrointestinal nematode infections. Mucosal Immunol. 11 (2), 304–315. https://doi.org/10.1038/mi.2017.113.

Sotillo, J., Sanchez-Flores, A., Cantacessi, C., Harcus, Y., Pickering, D., Bouchery, T., Camberis, M., Tang, S.-C., Giacomin, P., Mulvenna, J., Mitreva, M., Berriman, M., LeGros, G., Maizels, R.M., Loukas, A., 2014. Secreted proteomes of different developmental stages of the gastrointestinal nematode *Nippostrongylus brasiliensis*. Mol. Cell. Proteomics 13 (10), 2736–2751. https://doi.org/10.1074/mcp.M114.038950.

Stewart, C.J., 2023. Diet-microbe-host interaction in early life. Science 381 (6653), 36–38. https://doi.org/10.1126/science.adi6318.

Stryiński, R., Fiedorowicz, E., Mateos, J., Andronowska, A., Łopieńska-Biernat, E., Carrera, M., 2025. Exploring the exoproteome of the parasitic nematode *Anisakis simplex* (*s. s.*) and its impact on the human host—an in vitro cross-talk proteomic approach. Front. Immunol. 16, 1509984. https://doi.org/10.3389/fimmu.2025.1509984.

Sun, T., Chance, R.R., Graessley, W.W., Lohse, D.J., 2004. A study of the separation principle in size exclusion chromatography. Macromolecules 37 (11), 4304–4312. https://doi.org/10.1021/ma030586k.

Tabb, D.L., Vega-Montoto, L., Rudnick, P.A., Variyath, A.M., Ham, A.-J.L., Bunk, D.M., Kilpatrick, L.E., Billheimer, D.D., Blackman, R.K., Cardasis, H.L., Carr, S.A., Clauser, K.R., Jaffe, J.D., Kowalski, K.A., Neubert, T.A., Regnier, F.E., Schilling, B., Tegeler, T.J., Wang, M., et al., 2010. Repeatability and reproducibility in proteomic identifications by liquid chromatography-tandem mass spectrometry. J. Proteome Res. 9 (2), 761–776. https://doi.org/10.1021/pr9006365.

Tanaka, H., Sato, K., Saito, Y., Yamashita, T., Agoh, M., Okunishi, J., Tachikawa, E., Suzuki, K., 2003. Insect diapause-specific peptide from the leaf beetle has consensus with a putative iridovirus peptide. Peptides 24 (9), 1327–1333. https://doi.org/10.1016/j.peptides.2003.07.021.

Tang, S., Tao, J., Li, Y., 2024. Challenges and solutions for the downstream purification of therapeutic proteins. Antib. Ther. 7 (1), 1–12. https://doi.org/10.1093/abt/tbad028.

Tariq, K.A., 2015. A review of the epidemiology and control of gastrointestinal nematode infections of small ruminants. Proc. Natl. Acad. Sci. India B Biol. Sci. 85 (2), 693–703. https://doi.org/10.1007/s40011-014-0385-9.

Tarr, D.E.K., 2012a. Distribution and characteristics of ABFs, cecropins, nemapores, and lysozymes in nematodes. Dev. Comp. Immunol. 36 (3), 502–520. https://doi.org/10.1016/j.dci.2011.09.007.

Tarr, D.E.K., 2012b. Nematode antimicrobial peptides. Invertebr. Surviv. J. 9 (1). Article 1.

Tarr, D.E.K., 2016. Establishing a reference array for the CS-αβ superfamily of defensive peptides. BMC. Res. Notes 9 (1), 490. https://doi.org/10.1186/s13104-016-2291-0.

Torrent, M., Di Tommaso, P., Pulido, D., Nogués, M.V., Notredame, C., Boix, E., Andreu, D., 2012. AMPA: an automated web server for prediction of protein antimicrobial regions. Bioinformatics 28 (1), 130–131. https://doi.org/10.1093/bioinformatics/btr604.

Tran, T.D., Luallen, R.J., 2024. An organismal understanding of *C. elegans* innate immune responses, from pathogen recognition to multigenerational resistance. Semin. Cell Dev. Biol. 154, 77–84. https://doi.org/10.1016/j.semcdb.2023.03.005.

Tymecka, D., Misicka, A., 2020. Solution phase peptide synthesis: the case of Biphalin. In: Hussein, W.M., Skwarczynski, M., Toth, I. (Eds.), Peptide Synthesis: Methods and Protocols. Springer, US, pp. 1–11. https://doi.org/10.1007/978-1-0716-0227-0_1.

Ugolini, G.S., Wang, M., Secchi, E., Pioli, R., Ackermann, M., Stocker, R., 2024. Microfluidic approaches in microbial ecology. Lab Chip 24 (5), 1394–1418. https://doi.org/10.1039/D3LC00784G.

Ul Haq, I., Maryam, S., Shyntum, D.Y., Khan, T.A., Li, F., 2024. Exploring the frontiers of therapeutic breadth of antifungal peptides: a new avenue in antifungal drugs. J. Ind. Microbiol. Biotechnol. 51, kuae018. https://doi.org/10.1093/jimb/kuae018.

United Nations, 2024. Sustainable development goals: Take action for the sustainable development goals. Retrieved 16 May 2024, from. https://www.un.org/sustainabledevelopment/sustainable-development-goals/.

Uzoechi, S.C., Rosa, B.A., Singh, K.S., Choi, Y.-J., Bracken, B.K., Brindley, P.J., Townsend, R.R., Sprung, R., Zhan, B., Bottazzi, M.-E., Hawdon, J.M., Wong, Y., Loukas, A., Djuranovic, S., Mitreva, M., 2023. Excretory/secretory proteome of females and males of the hookworm *Ancylostoma ceylanicum*. Pathogens 12 (1), 95. https://doi.org/10.3390/pathogens12010095.

Van Bael, S., Zels, S., Boonen, K., Beets, I., Schoofs, L., Temmerman, L., 2018. A *Caenorhabditis elegans* mass spectrometric resource for neuropeptidomics. J. Am. Soc. Mass Spectrom. 29 (5), 879–889. https://doi.org/10.1007/s13361-017-1856-z.

Veltri, D., Kamath, U., Shehu, A., 2018. Deep learning improves antimicrobial peptide recognition. Bioinformatics 34 (16), 2740–2747. https://doi.org/10.1093/bioinformatics/bty179.

Vermeire, B., Gonzalez, L.M., Jansens, R.J.J., Cox, E., Devriendt, B., 2021. Porcine small intestinal organoids as a model to explore ETEC–host interactions in the gut. Vet. Res. 52 (1), 94. https://doi.org/10.1186/s13567-021-00961-7.

Wang, G., 2020. The antimicrobial peptide database provides a platform for decoding the design principles of naturally occurring antimicrobial peptides. Protein Sci. 29 (1), 8–18. https://doi.org/10.1002/pro.3702.

Wang, P., Robert, L., Pelletier, J., Dang, W.L., Taddei, F., Wright, A., Jun, S., 2010. Robust growth of *Escherichia coli*. Curr. Biol. 20 (12), 1099–1103. https://doi.org/10.1016/j.cub.2010.04.045.

Wang, G., Li, X., Wang, Z., 2016. APD3: the antimicrobial peptide database as a tool for research and education. Nucleic Acids Res. 44 (D1), D1087–D1093. https://doi.org/10.1093/nar/gkv1278.

Wang, B., Zhang, F.-W., Wang, W.-X., Zhao, Y.-Y., Sun, S.-Y., Yu, J.-H., Vitek, M.P., Li, G. F., Ma, R., Wang, S., Hu, Z., Chen, W., 2022a. Apolipoprotein E mimetic peptide

COG1410 combats pandrug-resistant *Acinetobacter baumannii*. Front. Microbiol. 13, 934765. https://doi.org/10.3389/fmicb.2022.934765.

Wang, G., Zietz, C.M., Mudgapalli, A., Wang, S., Wang, Z., 2022b. The evolution of the antimicrobial peptide database over 18 years: milestones and new features. Protein Sci. 31 (1), 92–106. https://doi.org/10.1002/pro.4185.

Wilbers, R.H.P., Westerhof, L.B., van Noort, K., Obieglo, K., Driessen, N.N., Everts, B., Gringhuis, S.I., Schramm, G., Goverse, A., Smant, G., Bakker, J., Smits, H.H., Yazdanbakhsh, M., Schots, A., Hokke, C.H., 2017. Production and glyco-engineering of immunomodulatory helminth glycoproteins in plants. Sci. Rep. 7, 45910. https://doi.org/10.1038/srep45910.

Wingfield, P.T., 2015. Overview of the purification of recombinant proteins. Curr. Protoc. Protein Sci. 80, 6.1.1–6.1.35. https://doi.org/10.1002/0471140864.ps0601s80.

Wong, M.T.J., Anuar, N.S., Noordin, R., Tye, G.J., 2023. Soil-transmitted helminthic vaccines: where are we now? Acta Trop. 239, 106796. https://doi.org/10.1016/j.actatropica.2022.106796.

Wright, E., Neethirajan, S., Weng, X., 2015. Microfluidic wound model for studying the behaviors of *Pseudomonas aeruginosa* in polymicrobial biofilms. Biotechnol. Bioeng. 112 (11), 2351–2359. https://doi.org/10.1002/bit.25651.

Xiao, X., Wang, P., Lin, W.-Z., Jia, J.-H., Chou, K.-C., 2013. iAMP-2L: A two-level multi-label classifier for identifying antimicrobial peptides and their functional types. Anal. Biochem. 436 (2), 168–177. https://doi.org/10.1016/j.ab.2013.01.019.

Xu, J., Li, Y., Wang, H., Zhu, M., Feng, W., Liang, G., 2021. Enhanced antibacterial and anti-biofilm activities of antimicrobial peptides modified silver nanoparticles. Int. J. Nanomedicine 16, 4831–4846. https://doi.org/10.2147/IJN.S315839.

Yan, K., Lv, H., Guo, Y., Peng, W., Liu, B., 2023. sAMPpred-GAT: prediction of antimicrobial peptide by graph attention network and predicted peptide structure. Bioinformatics 39 (1), btac715. https://doi.org/10.1093/bioinformatics/btac715.

Yang, Z., Choi, H., Weisshaar, J.C., 2018. Melittin-induced permeabilization, re-sealing, and re-permeabilization of *E. coli* membranes. Biophys. J. 114 (2), 368–379. https://doi.org/10.1016/j.bpj.2017.10.046.

Ying, M., Qiao, Y., Yu, L., 2016. Evolutionary expansion of nematode-specific glycine-rich secreted peptides. Gene 587 (1), 76–82. https://doi.org/10.1016/j.gene.2016.04.049.

Zachos, N.C., Kovbasnjuk, O., Foulke-Abel, J., In, J., Blutt, S.E., Jonge, H.R., de Estes, M. K., Donowitz, M., 2016. Human enteroids/colonoids and intestinal organoids functionally recapitulate normal intestinal physiology and pathophysiology. J. Biol. Chem. 291 (8), 3759–3766. https://doi.org/10.1074/jbc.R114.635995.

Zakharchenko, N.S., Rukavtsova, E.B., Gudkov, A.T., Bur'ianov, I.I., 2005. Enhanced resistance to phytopathogenic bacteria in transgenic tobacco plants with synthetic gene of antimicrobial peptide cecropin P1. Genetika 41 (11), 1445–1452.

Zakharchenko, N.S., Rukavtsova, E.B., Gudkov, A.T., Yukhmanova, A.A., Shkol'naya, L. A., Kado, C.I., Bur'yanov Ya, I., 2007. Expression of the artificial gene encoding antimicrobial peptide cecropin P1 increases the resistance of transgenic potato plants to potato blight and white rot. Dokl. Biol. Sci. 415 (1), 267–269. https://doi.org/10.1134/S0012496607040059.

Zhang, Y., Li, Z., Li, L., Rao, B., Ma, L., Wang, Y., 2021. A method for rapid screening, expression, and purification of antimicrobial peptides. Microorganisms 9 (9), 1858. https://doi.org/10.3390/microorganisms9091858.

Zhu, S., Gao, B., 2014. Nematode-derived drosomycin-type antifungal peptides provide evidence for plant-to-ecdysozoan horizontal transfer of a disease resistance gene. Nat. Commun. 5, 3154. https://doi.org/10.1038/ncomms4154.

Zhu, S., Gao, B., Tytgat, J., 2005. Phylogenetic distribution, functional epitopes and evolution of the CSαβ superfamily. Cell. Mol. Life Sci. 62 (19), 2257–2269. https://doi.org/10.1007/s00018-005-5200-6.

Zhu, Y., Mohapatra, S., Weisshaar, J.C., 2019. Rigidification of the *Escherichia coli* cytoplasm by the human antimicrobial peptide LL-37 revealed by superresolution fluorescence microscopy. Proc. Natl. Acad. Sci. 116 (3), 1017–1026. https://doi.org/10.1073/pnas.1814924116.